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Title: Cancer Induction by Repeated Doses of High-LET Radiation

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Scope of the Project

A main objective of the proposed work is to test the hypothesis that the enhancement of carcinogenesis seen for 2 exposures to high-LET radiation occurs equally after any number of exposures, up to 10. The more severe, less repairable type of DNA damaged caused by high in comparison to low-LET radiation could accelerate cancer progression by damaging the DNA of progressing cells. Rats were exposed to doses of the 56Fe beam at 1.1 geV/nucleon from the Brookhaven AGS either with or without dietary retinyl acetate. In conjunction with a cancer risk model, these data will help us understand how similar the 56Fe beam is to an argon ion beam and whether retinyl acetate is an inhibitor of 56Fe-induced cancer. Human and rat skin are highly susceptible to the carcinogenic action of ionizing radiation. Both develop a mixture of basal and squamous cell carcinomas with the rat exhibiting in addition a small incidence of sarcomas (1). Prevailing evidence indicates that a certain fraction of the initial molecular damage is converted to a variety of non-repairable alterations. Even so, cells harboring the relevant mutations do not necessarily express the cancer phenotype immediately and may do so only after many months or even years. An understanding of the differing biological effects of high and low-LET radiation is important for predicting the cancer risk of radiation exposure. High-LET radiation exposure results in a greater amount of initial DNA damage, unrepaired DNA strand breaks, and cell death than low-LET (2-4). The different features of high and low-LET radiation also correlate with differences in molecular events associated with progression of rat skin tumors (5).

Retinoids have shown chemopreventive activity in several animal models including irradiated rat skin. But the molecular mechanisms that lie behind the retinoid inhibition of radiation-induced cancer remain to be established. As an initial approach to a better understanding of this mechanism, this project focuses on discovering how a non-toxic, naturally occurring retinoid in the diet (retinyl acetate) alters gene expression upon exposure of rat skin to radiation. Specifically, the differential display method based on gene chip technology is used to identify genes whose expression is 1) altered by electron radiation alone, 2) or by retinyl acetate alone, 3) or by electron radiation but reversed by the presence of retinyl acetate. Furthermore, the affected genes will be grouped in accordance with their cellular functions in vivo, so as to suggest possible pathways that are involved in the process. Results from this work will increase our understanding of how retinyl acetate alters gene expression patterns in rat skin upon electron irradiation at carcinogenic doses. To the extent that retinoid-responsive genes are involved in carcinogenesis, the results may help to identify the genes involved in the broad mechanism of the retinoid's role in skin cancer, eg., proliferation-, differentiation-, or apoptosis-related genes.

2. Summary of Progress during 2002

Major progress has been made to define the gene expression level differences in rat skin exposed to electron radiation, dietary retinoid and the combination of radiation/retinoid in preparation for similar experiments utilizing the 56Fe beam. The goals of the proposed experiments are to identify genes whose expression is induced (or blocked) by radiation and is counter-acted by a specific retinoid, vitamin A, in the rat diet, and to reveal gene expression patterns that are affected by each treatments individually. This study would be the first in vivo study on this topic. It is inspired by the established facts that

radiation is a strong carcinogen, and that the presence of vitamin A can reduce its carcinogenic effects.

Although the role of gene expression changes in early stages of carcinogenesis are not fully understood, expression reversals may provide important clues regarding how the vitamin A might be acting. The differentially displayed genes could belong to differentiation or cell cycle genes, that impact cancer progression. The results will not necessarily make clear all of the genes that directly contribute to vitamin A inhibition of radiation carcinogenesis but should greatly narrow the list of possibilities.

The identification of genes whose expression is reversely altered by vitamin A after radiation, will add to our knowledge of the molecular events that underlie vitamin A's gene-regulating function on irradiated rat skin epithelium. This, in turn, will contribute to our understanding of the mechanism of chemoprevention of radiation-induced skin cancers by vitamin A. The steps are schematically shown below.

Rats on regular or vitamin A supplemented food

1. Irradiation
2. RNA extraction
3. reverse transcription to cDNA and transcription to biotin-labeled cRNA probe
4. hybridization to rat genome gene chip
5. data analysis
6. northern blot and RT-PCR to confirm some of the differentially displayed genes
7. categorize the patterns of gene expression alterations.

The general laboratory procedure of the proposed experiment is as follows: Upon arrival, eight 23 days old rats are divided into 2 dietary groups of 4 rats each: Group 1 be fed with regular food chow, Group 2 with food chow mixed with 250 ppm vitamin A acetate supplement. One week later, all the rats get 9 Gy or 18 Gy of electron radiation on two different dorsal patches about the size of 1.5cm x 2cm.

Twenty-four hours or 7 days after the irradiation, rats are killed, and their radiation exposed skin and non-irradiated skin are excised and pooled according to their radiation dose and vitamin A exposure. Total RNAs are extracted with TriZol reagent (Gibco BRL, NY). The total RNAs are reverse transcribed to prepare double strand cDNA by using the dT23 primer adjacent to T-7 RNA polymerase promoter region, double stranded cDNAs then are transcribed into biotin-labeled cRNA through *in vitro* transcription (Enzo Diagnostics, NY), as suggested by the gene chip manufacturer.

The cRNAs are differentially displayed by means of Affymetrix rat genome gene chips (RAT U34-A) on Affymetrix chip system (Affymetrix, CA). Each chip contains 8740 specific probes for rat genes, including 1783 sequence tags plus about 6957 known genes representing every known functional category. With the gene chip, expression of all 8740 genes and gene groups can be examined in one single hybridization. Briefly, the procedure is as following:

1). Eight 3-week old rats are equally divided into 2 dietary groups: 1) on regular chow for 7 days before and 1 day post irradiation, 2) 250 ppm vitamin A in chow, for 7 days before and 1 day post-irradiation.

2) Radiation treatments: each rat gets 0, 9 or 18 Gy radiation on 3 different areas (each 1.5 cm x 2 cm) on their shaved dorsal skin, thus making 6 different treatments (3 radiation doses and 2 dietary groups).

3) Control and radiation exposed skin are excised from the animals 24 hours or 7 days after the irradiation and pooled from 4 rats for each treatment. The skin is immediately immersed in TriZol and homogenized. Then stored in -70oC for total RNA extraction.

4) Total RNA is reverse transcribed into cDNA and then made into cDNA. The cDNA in turn is used to make biotin-labeled cRNA through in vitro transcription, and then the cRNA is hybridized to the probes on gene chips. Chip analysis was not used for 18Gy + vitamin A treatment, leaving 5 groups to be analyzed for gene expression.

Five gene chip hybridizations have been carried out for control diet rat samples with 0, 9 Gy and 18 Gy radiation (designated as control, 9Gy and 18Gy respectively), and vitamin A treated rat samples with 0 and 9 Gy radiation (Vitamin only, Vitamin + 9Gy). The expression level for each gene was normalized across different chips by adjusting the total levels of all 5 chips to the total in the control sample. Expressions below 0 (after background subtraction) were considered to be 0. The resulting expression signals ranged from 0 to 278,710 (for EST228160), with an average of 4217. If only signals above 1000 are considered significant, then 4402 of the 8740 (50.4%) genes are positive in at least one of the 5 groups, and the remaining 4338 genes (49.6%) are considered as not significantly expressed in any of the 5 chips. Since genes with lower expression levels tend to have larger variations, the following analysis uses a cut-off line of ≥1000 intensity to distinguish positively from un-expressed genes.

By comparing two chips side by side with signal strength of ≥1000 in at least one sample, the results were achieved by counting the number of genes with complete induction/inhibition or with ≥5 fold change: Nine Gy radiation (9Gy) vs. control (control) showed 188 altered genes; 18 Gy radiation (18Gy) vs. control showed 92 altered genes, among which 54 genes were also altered by 9Gy vs. control, showing a radiation dose-response; 18 Gy vs. 9 Gy radiation (18Gy vs. 9Gy) showed 77 altered genes; vitamin A treatment alone (vitamin only) vs. control showed 231altered genes; 9 Gy radiation vs. 9 Gy radiation plus vitamin A (9Gy vs. vitamin + 9Gy) treatment showed 58 altered genes.

One approach to study genes of interest is to find the genes that are reversely regulated by vitamin A and radiation and show no discrepancy across all five chips, that is, genes that are responsive to both vitamin A and radiation in opposite ways. There are 56 genes that show at least 2.5 fold change between 9 Gy (9Gy) and 9 Gy plus vitamin A (vitamin + 9Gy) chips, and also have consistency in other chips. Among these genes, different expression patterns were found as listed in Table 1. In one group, genes were repressed by vitamin A alone and were induced by radiation, and the radiation induction was reduced by the retinoid. In another group, vitamin A alone did not change the expression in comparison to control control, but the combination of vitamin A and radiation reverted the radiation induction back to control levels. Other genes, such as,

glycam I were inhibited by radiation, but were reversed to control level by vitamin A (details in Table 1). Among the genes that showed the greatest reversal of radiation effects by vitamin A treatment are: keratin 14 (K14), ornithine decarboxylase (ODC), an MRP-14 (intracellular calcium-binding protein gene), and an EST rc_AA957003 which is similar to intracellular calcium-binding protein MRP 8.

Table 1 Genes showing expression ratios >= 5 between radiation and vitamin A+radiation♦

	co	vo*	c9*	v9*	c18*	Gene Descriptions
Group 1. vitamin A reverses radiationinduction						
G1-A. vitamin A alone does not alter the gene expression						
1	4758	4162	55945	8190	21710	D63774 RATRETK rat n. mRNA for keratin 14
2	42	593	18930	1250	24441	L18948 intracellular calcium-binding protein (MRP14)
3	4911	3165	17865	3386	21593	rc_AA957003 UI-R-E1-fq-d-09-0-UI.s1 3 end
4	4393	3719	7578	3019	6448	X71127 complement protein C1q beta chain
5	0	0	4060	0	4038	AF068861 rat n. beta defensin-2
6	0	54	2424	0	908	X82669completeSeq R.norvegicus RT1.Au gene
7	871	278	2380	604	1326	J05495 Rat Gal/GalNAc-specific lectin
8	0	0	2298	347	4011	X13098cds RNURICAS urate oxidase (uricase)
9	47	286	2031	0	3653	AF017393 cytochrome P4502F4 (CYP4502F4)
10	179	45	1993	377	1433	A30543 p-Meta-1for CD44 surface protein
11	151	98	1792	0	76	L31546 RAT5HT2X Rat serotonin 5HT-2 receptor
12	426	642	1632	621	1397	rc_AA799861 EST189358 rat n. cDNA, 3 end
13	109	720	1069	0	180	J05405 RATHO2 Rat heme oxygenase-2 (HO2)
14	0	0	1013	0	755	M17523 RATGHYY peptide tyrosine-tyrosine(YY)
15	0	0	949	0	1299	rc_AA799691 EST189188 rat n. cDNA, 3 end
G1-B. vitamin A alone also inhibits the gene expression						
16	3232	931	6404	1366	2675	X14254 MHC class II-associated invariant chain
17	1400	418	3888	809	3288	X71127 complement protein C1q beta chain
Group 2. vitamin A reverses radiation inhibition						
18	1068	1191	30	1375	0	L08100 Rat glycam 1
19	830	1039	261	651	552	M64862 Rat matrin F/G
20	715	1348	50	1056	647	S81584 transforming GF-beta R kinase homolog
Group 3. vitamin A action is almost independent of radiation actions						
G3-A. vitamin A increases the gene expression						
21	1848	5388	3232	8916	3317	L28818cds RATINVO rat n. involucrin gene
22	1709	5537	3278	9482	4161	K02814 major acute phase alpha-1 protein (MAP)
23	1629	8498	2740	11666	5577	K02814 major acute phase alpha-1 protein (MAP)
24	538	3192	1020	2957	1298	rc_AA858607 UI-R-E0-bq-a-08-0-UI.s1cDNA
25	156	1149	143	1038	395	rc_AI009771 EST204222 rat n. cDNA, 3 end
26	61	2117	230	1600	199	rc_AA891664 EST195467 rat n. cDNA, 3 end
27	0	1544	28	762	0	rc_AA859633 UI-R-E0-bs-h-09-0-UI.s1 3 end

G3-B. vitamin A decreases the gene expression						
28	5582	2082	4369	1765	9054	U92081 epithelial cell transmembrane protein precursor
29	4037	1990	3700	1228	4569	J04792 ornithine decarboxylase (ODC) gene
30	3800	900	4683	1295	3266	MHC class II A-beta RT1.B-b-beta gene
31	3631	1704	3487	1338	2870	U49694 cytosolic acyl co-A thioester hydrolase
32	2771	881	1435	525	2359	mixed-tissue library rat n. cDNA clone rx04769
33	1952	207	1054	115	1118	rc_AA892346 EST196149 rat n. cDNA, 3 end
34	1730	734	1887	567	1303	M57276 Rat leukocyte antigen MRC-OX44
35	1353	339	937	356	900	rc_AA799473 EST188970 rat n. cDNA, 3 end
36	1326	165	1339	246	572	rc_AI230130 EST226825 rat n. cDNA, 3 end
37	1129	0	921	332	1073	rc_AA859661 UI-R-E0-bs-c-04-0-UI.s1, 3 end
38	1049	275	1383	389	1301	Rat mixed-tissue library cDNA clone rx02866 3
39	1028	0	809	298	677	Rat mixed-tissue library cDNA clone rx01964 3
40	969	662	1512	522	1437	rc_AA875227 UI-R-E0-cq-a-09-0-UI.s1 3 end
41	905	442	1148	174	1641	M65149 Rat CELF
42	811	186	958	232	1804	U36992 cytochrome P450 Cyp7b1
43	806	0	1403	406	2196	AB011531 rat n. mRNA for MEGF5,
44	746	174	1113	313	1300	AF081144 rat n. CL1AA
45	742	0	1032	262	804	rc_AA892849 EST196652 rat n. cDNA, 3 end
46	680	0	1029	0	911	U02553 protein tyrosine phosphatase gene,
47	511	0	627	223	1441	U68272 RNU68272 interferon gamma receptor
48	487	0	1017	0	1174	rc_AA893596 EST197399 rat n. cDNA, 3 end
Group 4. vitamin A and radiation have additive or synergistic effects						
49	21726	3625	12395	2802	12360	X83537 membrane-type matrix metalloproteinase
50	5782	2744	3354	1078	5456	rc_AA892895 EST196698 rat n. cDNA, 3 end
51	4676	2466	3283	453	5191	rc_AA799328 EST188825 rat n. cDNA, 3 end
52	4594	0	1574	530	1788	AA685376 ~ to NADH-ubiquinone oxidoreductase
53	2928	851	1178	394	1364	L15556 rat n. phospholipase C (BETA4)
54	1897	554	900	14	1387	D82868 Rat mRNA for bHLH protein
55	318	2262	1280	4954	319	Rat mixed-tissue library clone rz00721 3 end

♦: Rad, radiation; vitamin A, retinyl acetate

*: samples were normalized to the total sum of intensities in control. 9Gy, 18Gy, control diet but 9 or 18 Gy Rad; vitamin only, vitamin + 9Gy, vitamin A diet with 0 or 9 Gy Rad.

Another goal of the study is to find the most affected functional groups of genes. To do this, genes known to have certain functions are listed together, such as oncogene/tumor suppressor genes, cell cycle, apoptosis, differentiation, and redox genes, etc. Genes in the functional group are checked to see if any show significant change (>2 fold). It turned out several groups failed to show any genes with significant changes after radiation with/without vitamin A treatments, including apoptosis-related genes and oncogene/tumor suppressor genes. Among the 16 cytokines on the chip, only one (Cyclin G) showed >2 fold change between 9Gy and vitamin + 9Gy.

To further study how radiation and vitamin A alone can affect the rat skin gene expression, each of the most greatly affected 100 genes by 9 Gy radiation alone (9Gy vs.

control) or by vitamin A alone (vitamin only vs. control) were fitted into one functional category, then the number of genes in each category were counted. As control, 100 randomly picked up genes from expressed control sample were also assigned into different functional categories (see Figure 1).

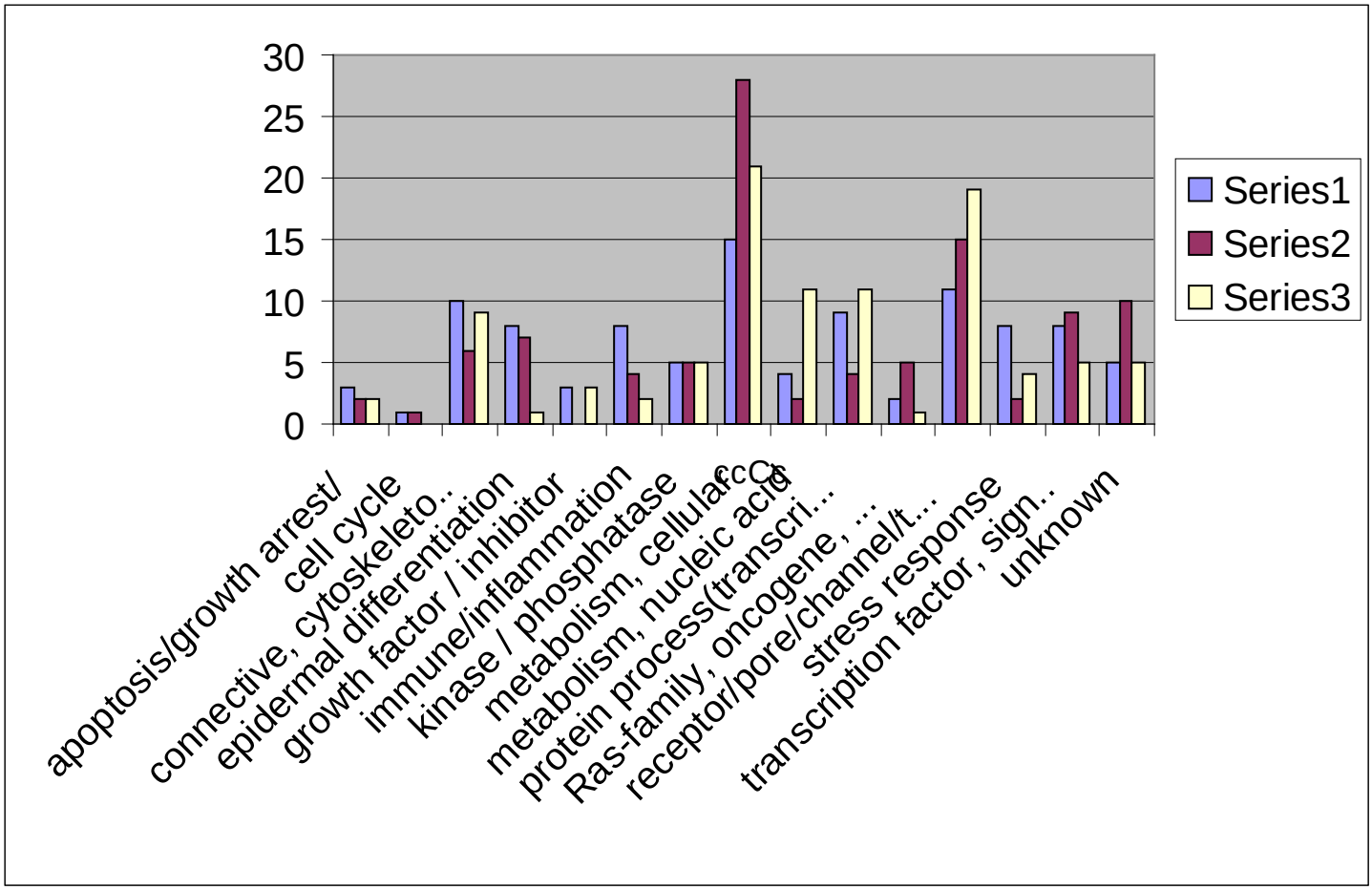


Figure 1. Comparison of the functional groups that were affected by radiation and vitamin A. The 100 most altered genes by 9 Gy radiation or vitamin A versus no treatment control control were listed and assigned into one functional group, 100 randomly chosen genes from the expressed control gene list were also included in the chart as control.

The chart shows that the most affected genes by 9 Gy or vitamin A belong to the category of differentiation, stress response, inflammation/immune related, and nucleic acid metabolism. Northern blots showed a similar trend of expression as the gene chip for keratin 14 gene, a putative marker of proliferating basal cells, although expression ratios differed. For example, on the gene chip K14 showed 11.7 fold increase after 9 Gy (9Gy) over control (control) and vitamin A treatment (vitamin + 9Gy) reversed the expression back to basal level in control; while on northern blots the increase in 9Gy over control was 4.9 fold, and vitamin A treatment (vitamin + 9Gy) reduced the expression to 2.2 fold above basal level in control. The vitamin A reversible increase after 9 or 18 Gy radiation suggests

that vitamin A may inhibit the dedifferentiating effects of radiation, thus restoring the normal differentiation status of the epidermis. Real-time PCR was also used to confirm the gene chip results. So far, the gene chip expression pattern of the involucrin gene has been confirmed by reverse-transcription PCR on the Light Cycler (La-Roche, IN).

The expression of many in vivo rat skin genes were found to be altered by radiation or vitamin A treatments, and vitamin A reversed some radiation effects on gene expression. Functional analysis of the data implies the active involvement of differentiation, stress response, immune/inflammation, and nucleic acid metabolism in response to radiation or vitamin A. Further more, several differentiation-related and calcium response genes showed vitamin A reversal of the radiation effects on the gene expression, suggesting differentiation and calcium pathway may play an important role in vitamin A inhibition of radiation effects on rat skin model. More detailed analysis on the gene chip data for lower fold alteration will reveal more patterns of gene expression alterations, leading to other possible pathways that might be relevant to vitamin A's reversal of radiation action.

Carcinogenesis Experiments

Rat skin was exposed to the plateau region of the 1.0 geV/nucleon 56Fe beam at the Brookhaven AGS. The rats were exposed 3 at a time in an apparatus that allowed about 8.0 cm^2 of skin from each animal to be folded up into the beam while the remainder of the body was shielded by 33 cm of lucite. After irradiation the animals were observed weekly for the initial 4 weeks then at 6 week intervals thereafter. Current data was obtained based on a follow-up time of 54 weeks. Eventually the follow-up time will be about 100 weeks. All lesions were noted, photographed and identified for eventual histological diagnosis. The data shown below have not been verified histologically but are believed to be about 90% accurate based previous experience. Error bars were estimated from standard errors which included a correction for quantized data.

Rat skin was exposed to the plateau region of the 1.0 geV/nucleon 56Fe beam at the Brookhaven AGS. The rats were exposed 3 at a time in an apparatus that allowed about 6.0 cm2 of skin from each animal to be folded up into the beam while the remainder of the body was shielded by 33 cm of lucite. After irradiation the animals were observed weekly for the initial 4 weeks then at 6-week intervals thereafter. Current data was obtained based on a follow-up time of 48 to 68 weeks. Eventually the follow-up time will be about 100 weeks. All lesions were noted, photographed and identified for eventual histological diagnosis. Error bars were estimated from a standard error calculation that included a correction for quantized data.

The protocol of the 1.1 geV/nucleon 56Fe exposures is shown in Table 2. The data show that single doses of 56Fe ions are somewhat more effective than argon in producing tumors at 4.5 Gy but are about equally effective at 3.0 Gy and 9.0 Gy. The evidence so far for a split dose effect is inconclusive. However a strong tendency is apparent for the tumors to be inhibited by the presence of 250 ppm retinyl acetate in the lab chow diet. These are data represent only about one-half the eventual number of tumors expected. Even so the tumor yields are not likely to change substantially, although error bars are expected to narrow with increasing total tumor numbers.

Table 2. Protocol and Summary of Results – Fibroma Induction by 56Fe ions

Group	Dose	# of Rats			Dose Pattern	Total Tumors	Cumulative Incidence +- S.E. (Tumors/rat)*
		Part I	Part II	Total			
1	3.0	7	9	16	Split	39	0.964 +- 0.669
2	3.0	15	17	32	Split, Vit. A	11	0.377 +- 0.196
3	9.0	5	9	14	Split	13	0.754 +- 0.652
4	9.0	18	18	36	Split, Vit. A	15	1.238 +- 0.318
5	4.5	0	9	9	Single	20	2.671 +- 1.070
6	4.5	0	9	9	Single, Vit. A	5	0.771 +- 0.429
7	9.0	7	9	16	Single	23	1.431 +- .591
8	9.0	14	18	32	Single, Vit. A	15	0.848 +- 0.627
9	3.0	3	9	12	Single	42	1.199 +- 0.439
10	3.0	12	15	27	Single, Vit. A	22	0.612 +- 0.144
Totals		81	122	203		205	

*at 1 year post-exposure

The oncogenic action of multiple doses of high-LET 56Fe ion radiation (average LET=125 keV/μ) in comparison to single doses of argon ion radiation was examined in the rat skin. Data from 2 fractions of argon ion radiation suggest that multiple exposures produce more carcinomas than expected from the summation of single dose responses. Moreover low-LET induced carcinomas were 90% preventable by dietary retinyl acetate (RA) which is a relatively non-toxic, naturally occurring retinoid known to have strong effects on cell proliferation, gene expression and differentiation in keratinocytes. The new findings reported here indicate that high-LET induced fibromas may be as responsive to the inhibitory effects of RA as low-LET carcinomas were in previous studies, but there probably is no split dose enhancement as seen for carcinomas induced by argon ion radiation.

A second goal of these experiments to obtain information related to the mechanism of retinyl acetate inhibition of skin cancers. Previous studies have indicated that about 90% of cancers induced by single doses of low-LET electron radiation in rat skin can be prevented by 250 ppm dietary retinyl acetate. The approach here is to focus on radiation inducible genes that are retinoid responsive. These are the genes most likely to be involved in the cancer inhibitory effect of the retinoid. Their identification will improve our understanding of which genes may be contributory to retinoid-based cancer prevention in rat skin and will establish how gene inhibition patterns depend on the LET and dose rate of the radiation.

Retinoids have been tested for their chemoprevention activity in several animal models including irradiated rat skin. The molecular mechanisms that lie behind the retinoid inhibition of radiation-induced cancer remain to be established. As an initial approach to a better understanding of this mechanism, this project is attempting to discover how dietary retinyl acetate alters gene expression upon exposure of rat skin to electron-irradiation. The

differential display method was used to identify genes whose expression is induced by electron radiation but was inhibited (or enhanced) by the presence of retinyl acetate. To the extent that retinoid-responsive genes would be involved in carcinogenesis, the results will help to explain the retinoid's role in cancer inhibition.

An experiment entitled, "The Effects of a Dietary Retinoid on Gene Expression in Skin Following 56Fe Irradiation" was irradiated during the January 2002, NASA beam run on the AGS. Preliminary findings with low LET radiation indicate that dietary retinyl acetate counteracts radiation-induced expression in hundreds of genes. Several of the quantitatively most affected genes are known markers of cell proliferation in keratinocytes raising the possibility that the retinoid acts by preventing radiation-induced increases of cell proliferation genes. There is no information available on the effect of high LET radiation, such as 56Fe at the AGS, on gene expression in skin keratinocytes. The purpose of the new experiment was to obtain information on gene expression patterns following skin exposure to the 56Fe beam in rats on regular lab chow or retinyl acetate supplemented lab chow. Results of earlier experiments at the AGS over the past 3 years indicate that the dietary retinoid is almost as effective against induction of malignant and benign skin tumors induced by 56Fe as found previously for low LET electron radiation. Rats were exposed in the same manner as previously in carcinogenesis experiments. At 24 hours or 7 days after exposure the animals were killed and the irradiated tissue was excised and frozen in liquid nitrogen for later analysis of RNA expression patterns by making use of the Affymetrix gene chip technology.

These experiments will provide evidence of possible similarities and differences between high and low LET radiation on changes in gene expression patterns. It will also help to delineate the mechanism by which dietary retinoids are able to prevent about 60% of the tumors that might otherwise occur following irradiation with 56Fe. Gene expression patterns will be assayed on Affymetrix rat chips containing 8740 known and unknown genes. Rats will be grouped as follows: controls=no treatment, vitamin A=dietary retinoid only, vitamin A + 3 Gy 56Fe, vitamin A+1.5 Gy 56Fe, rad3=3 Gy 56Fe only, and rad1.5=1.5 Gy only. Three rats will be in each group for a total of 15 rats only 12 of which will be irradiated. Heavy ions are needed to simulate the space environment that is under study in these NASA-supported experiments.

Continuation of carcinogenesis experiments:

In previous studies rat skin was exposed to the plateau region of the 1.0 geV/nucleon 56Fe beam at the Brookhaven AGS. Approximately 180 rats were exposed 3 at a time in an apparatus that allowed about 6.0 cm2 of skin from each animal to be folded up into the beam while the remainder of the body was shielded by 33 cm of lucite. After irradiation the animals were observed weekly for the initial 4 weeks then at 6-week intervals thereafter. In some groups 250 ppm retinyl acetate was continuously available the lab chow beginning 1 week before irradiation. Current data was based on a follow-up time of 80 weeks. The preponderance of the tumors so far are probably fibromas. These are the earliest and most frequent tumors and are useful for comparing the effects of different radiation exposures. The data show that single doses of 56Fe ions are somewhat more effective than argon in producing tumors at 4.5 Gy but are about equally effective at 3.0 Gy and 9.0 Gy. The presence of 250 ppm retinyl acetate in the lab chow diet reduced the incidence of tumors to about 30% of the incidence in groups exposed only to the radiation.

Interactions between electron radiation and vitamin A based on data from the gene expression microarray study suggest that vitamin A can inhibit electron radiation-induced proliferation and inflammation responses, and some fibroblast or macrophage migration/invasion responses as well. Furthermore, vitamin A promoted epidermal differentiation on top of the electron radiation-stimulated abnormal epidermal differentiation, thus possibly protecting the skin from cancer progression by promoting the death of potentially electron radiation damaged cells. vitamin A promoted epidermal differentiation on top of the electron radiation induced abnormal differentiation. In some ways, terminal differentiation of epidermal cells act in parallel to apoptosis in preventing the damaged cells from producing more cells. The ability to stimulate epidermal differentiation could render vitamin A the chemopreventive properties. Together with the inhibition of proliferation, these vitamin A effects have the potential to limit the number of electron radiation-damaged cells to pass the damages to their descendants, thus beneficial to the organism eventually. Results from the present study are consistent at the gene expression level with prior speculations about the possible mechanism of retinoid's inhibition of skin cancer promotion and progression, that is, the retinoid was actively blocking the expression of genes associated with increased proliferation and inflammation following irradiation and accelerated the expression of genes associated with epidermal differentiation.

Ionizing radiation (including high and low linear energy transfer radiation - LET) is a well-known risk factor of cancer development; human and rat skin are highly susceptible to its carcinogenic action (6). When ionizing radiation passes through cells, it removes electrons from molecules in its track, which initiates a cascade of events that eventually result in DNA alterations some of which lead to permanent mutations. Direct DNA damages induced by IR are often clustered and contain abasic sites, oxidized purines, or oxidized pyrimidines (7, 8), resulting in cell death, chromosomal and DNA single strand breaks, generation of reactive oxygen moieties, as well as a complex array of cellular responses, including cell cycle arrest, DNA repair and/or apoptosis (9). Besides DNA damage, ionizing radiation can also cause infrequent genomic mutations through irradiation targeted to the cytoplasm (10) or through by-stander effects on neighboring cells adjacent to irradiated cells (11). The damage could lead to long-term destabilization of irradiated cells, which may lead to a further increase in mutation rate in the progeny of the exposed cells, a phenomenon called genomic instability (12, 13). Functionally ionizing radiation acts partly like an oxidant, since the addition of anti-oxidants partially eliminate radiation induced cellular damage, such as, generation of reactive oxidative species and lipid oxidation (14-16).

Mammalian cells have evolved several mechanisms for responding to damage created by radiation, such as, the activation or suppression of the transcriptional machinery, or the up-regulation of enzymes that are necessary for DNA repair or apoptosis (17). Furthermore, radiation is also capable of inducing a stress response in skin, which is accompanied by inflammatory processes controlled partly by pro-inflammatory cytokines (18). It's known that human cells respond to X-ray with inducible early, intermediate, and late responses at the transcriptional level (19). There are many reports on radiation inducible genes in mammalian cells or tissues, such as *P21 WAF1/Cip*1 (20, 21), c-*MYC*, *NEUROLEUKIN*, Co.Zn-*SOD*, *BCL*-2 (22). Some other genes are reported to be down-regulated by radiation in other cell lines or tissues, such as DNA *LIGASE I, CD19* (23),

CYCLIN B (24), *STATHMIN* (25), and c-*MYC*, *TOP II* (26). Thus radiation induced gene expression is seen to vary widely in cell lines of different origin and in different tissues, highlighting the importance of cellular context to genotoxic stress responses.

It's believed that many of the cellular responses to ionizing radiation are predominantly beneficial and protective to the response of the radiation damage, such as apoptosis, DNA repair, cell cycle arrest to allow DNA repair, terminal differentiation to stop the cells from dividing, proliferation to replace the lost cells, and inflammation to stimulate lymphocytes to remove the damaged cells, etc.. However, if the responses are inefficient or overwhelmed, the accumulated errors may cause a progressive instability eventually leading to neoplasias and cancer.

The prevailing theory to explain radiation carcinogenesis is a version of the more general multiple-stage theory. The latter hypothesis specifies that cells progress to cancer through a series of discrete, irreversible genetic alterations, however data on radiation-induced cancer incidence in rat skin suggests that at lower LET values an intermediate repairable alteration may occur prior to the formation of an irreversible mutation (6, 27). Carcinogen-induced mutations are frequently found in oncogenes and tumor suppressor genes in neoplastic tissue. Ionizing radiation is capable of activating oncogenes by a point mutation or by a chromosome translocation that transfers a gene to an inappropriate promoter region (28). Activated oncogenes and inactivated tumor suppressor genes have been seen during the late stage of radiation-induced cancers (29-31). For example, K-*ras* and/or c-*myc* oncogenes were activated in all tested rat skin tumors induced by low-LET electron radiation (32). Amplification of other oncogenes, such as c- *jun* and c-*fos*, in skin cancers have been reported elsewhere (33-35).

2. Rat Skin Model

Skin is composed of several layers including epidermis and dermis with a basal lamina separating them. The epidermis is a stratified epithelium characterized by a continuous replacement of the most superficial elements by proliferation and upward migration of the deeper layer, i.e., basal cells. Most skin cancers originate from either the epidermis or the epidermal appendages, e.g., hair follicles and sebaceous glands, all of which comprise the dermal epithelium. Many reagents, eg. retinoids, can profoundly alter the differentiation pattern of epidermis, even changing keratinizing cells into serous secreting cells (36). Skin is also one of the most important retinoid storage organs for the whole body (37).

Due to the distinctiveness of skin as the first line to defend the organism against outside toxicants, skin cells have evolved a unique form of terminal differentiation characterized by the formation of a cornified envelope to protect the cells beneath the surface. The cornified envelope consists of a stabilized array of keratin filaments contained within a covalently cross-linked protein envelope (38). The transition from the granular cells to the cornified cells involves various post-translational modifications that include deimination of arginine residues. Major deiminated epidermal proteins are derived from keratin I. Finally, the cornified cells are dead, flattened polyhedrons that are uniquely adapted to provide a moderately impervious, protective surface. Many internal and external stimuli can affect the normal differentiation process of the skin, because differentiation is closely tied to proliferation of the basal layer.

Even though skin epithelial cells generally differentiate toward dead cells, under some unusual stress circumstances, they still have the potential to de-differentiate and become purely proliferating cells. This is also true in other organs, for example, hepato-epithelial cells can de-differentiate into proliferating status when the organ has undergone mechanical injury (39).

The present study focuses on the rat skin as the target tissue. Rat skin was chosen because it is a frequently studied model in radiation carcinogenesis. In addition, it's more sensitive to radiation per cell than mouse skin and has more cells at risk (40). The use of rat skin has the advantage over an *in vitro* keratinocyte model, since the latter could not exactly reflect certain biological responses, such as, terminal differentiation or inflammation or changes in blood flow.

3. Retinoid's Biological Effects

Retinoids are the chemical derivatives of vitamin A. There are more than five thousand retinoid compounds of natural and synthetic origin (37); the majority of them are strong inhibitors of cell growth and promoters of differentiation (41). Retinoids have wide variety of effects in embryogenesis, reproduction, vision and regulation of inflammation, growth and differentiation of normal and neoplastic cells in vertebrates. Retinoids exert their activity by modulating gene expression following interaction with and activation of the nuclear retinoic acid receptors and transcription factors RAR and RXR (42). Each receptor has three isoforms: α, β and γ. RARs heterodimerize with RXRs and mediate retinoid activity, while RXRs could also heterodimerize with other members of the steroid receptor superfamily (43), making retinoid signaling highly complex, since multiple regulation pathways may be involved.

Retinoids have considerable anti-tumor activity against a variety of tumors, including mammary, head and neck, and skin tumors, and they have been put into clinical use for chemoprevention and chemotherapy of various tumors. But the mechanism of how retinoids inhibit tumor formation is not clear yet. So far they are proposed to induce apoptosis or act as antioxidants, inhibit cellular proliferation, or induce differentiation as shown by retinoic acid's therapeutic effects on acute promyelocytic leukemia.

4. Chemoprevention of Ionizing Radiation-Induced Cancer

There are many studies on the chemoprevention of cancers but very few on chemoprevention of ionizing radiation induced cancers. Research showed inhibition of ionizing radiation damage to plasmid DNA by the application of caffeine or chlorophyllin and that melatonin could prevent ionizing radiation plus dimethylbenz(a)anthracene induced rat mammary cancer (44).

Consistent results were achieved in our lab at New York University regarding chemoprevention of rat skin cancers induced by ionizing radiation. Previous work showed that, 90% of the rat skin cancers induced by single doses of 12 Gy low-LET radiation were prevented with 250 ppm dietary retinyl acetate (vitamin A acetate, vitamin A hereafter) intake when the vitamin A was given for life beginning 2 weeks before the radiation exposure, in comparison to a radiation-only group. It also showed the group receiving vitamin A beginning 2 weeks prior to radiation and ending 7 days after the radiation exhibited about 50% reduction in cancer incidence. These experiments suggest that vitamin

A can inhibit cancer-relevant event(s) that occur at about the time of radiation, and can counteract cancer-relevant events that occur late in the carcinogenic process as well. Thus it is possible that prevention of premutagenic, repairable alterations may play a role in prevention of radiation carcinogenesis by dietary retinoids.

Although there are numerous studies on retinoid or radiation effects on various kind of tissues and cell lines, so far, no literature has been found to report on gene expression alterations after retinoid and irradiation on *in vivo* skin. Therefore, it is essential to conduct an *in vivo* experiment to discover the true scenario. Since data on prevention of skin cancer by vitamin A in rats is already available, this compound was utilized as the test retinoid in this study. Electron radiation (LET = 0.34 kev/μ; electron radiation hereafter) was chosen because it contains the lowest linear energy transfer of all ionizing radiation types, and the penetration depth of it can be easily controlled and focused on the epidermis.

5. Rationale of the Experiment

The goals of this experiment are 1) to identify genes whose expression is induced (or blocked) by radiation and is counter-acted by a specific retinoid, RA, given at a non-toxic concentration in the diet, and 2) to reveal gene expression profiles that are affected by each individual treatment. This study is the first *in vivo* study on this topic. It is inspired by the established facts that radiation is a strong carcinogen, and vitamin A can reduce radiation-caused cancers in rat skin.

As discussed above, many reports on radiation and especially on retinoid effects were controversial due to the diverse *in vitro* conditions; even *in vivo* studies could vary greatly due to toxicity or the ways the retinoids were given (topically or by ingestion). Therefore, an *in vivo* study with minimal toxicity was deemed to be essential to reveal a more complete and representative picture of what really happens *in vivo*.

Although gene expression changes in tissues are complex and not fully understood, some reversal effects during the early phase of carcinogenesis may provide important information regarding how the vitamin A might be counteracting the early effects of ionizing radiation on skin. Although all differentially displayed genes are not necessarily relevant to cancer, the results have the potential for identifying subpopulations of genes as likely candidates. In fact the list of possibilities could be greatly narrowed out of the several thousand present on currently available rat expression chips.

The identification of genes whose expression is reversely altered by vitamin A after electron radiation, will add to our knowledge of the molecular events that underlie vitamin A's gene-regulating function on irradiated rat skin epithelium. This, in turn, will contribute to our understanding of the mechanism of chemoprevention of radiation-induced skin cancers by vitamin A. Thanks to the emergence of oligonucleotide microarray (gene chip) technology, it is now possible to screen thousands of gene expressions in a single hybridization, in contrast to the older methods which study only one or a few genes at a time. Even a few years ago, as we first started this gene expression study in 1997, the best technology was macro-arrays with hundreds of cDNA probes fixed on a nylon membrane. We have data showing that the reproducibility and stability of those macro-arrays were not comparable to the present-day expression microarrays.

Materials & Methods:

The overall procedure of this study is schematically shown.

rats fed regular food *rats fed food with vitamin A supplement*

⇓ for 7 days prior to irradiation

Low LET electron irradiation at time zero (0, 9 or 18 Gy)

⇓

RNA extraction at 1 day and 7 days after irradiation

⇓

reverse transcribe to cDNA and biotin-label

⇓

hybridize to rat genome gene chip data analysis

⇓

perform northern blot or RT-PCR to confirm differentially displayed genes

⇓

categorize alterations of the gene expression patterns

Animals and Treatments:

Male Sprague-Dawley rats were delivered at 21 days of age from Charles River Laboratory (Charles River Labs, MA) and kept in the Animal Facility of the Nelson Institute under standard conditions, i.e. food and water *ad libitum*, light/dark – 12 hours/12 hours, 15 air changes per hour, etc. The rats were divided into dietary groups: Group 1 (8 rats) were immediately fed with regular lab chow (Purina, NJ), Group 2 (8 rats) or with lab chow mixed with 250 ppm retinyl acetate (Roche Inc., NJ). Seven days later (defined as time zero), the rats were irradiated on 3 separate shaved dorsal skin patches (about 2cm x 2cm in area) with either 0 Gy, 9 Gy or 18 Gy of electrons, as shown in Figure 2. For irradiation the rats were away from the home cages for about 30 minutes, after which they were returned to their home cages and their diets were resumed. For restraint during irradiation the rats were anesthetized with 35 mg/kg sodium pentobarbital i.p. At 1 day and 7 days, 4 rats from each dietary vitamin A regimen were killed, and dorsal skin patches from the 3 irradiated areas were collected.

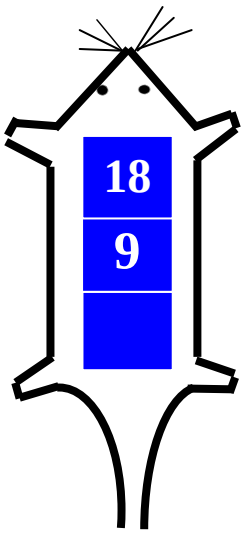
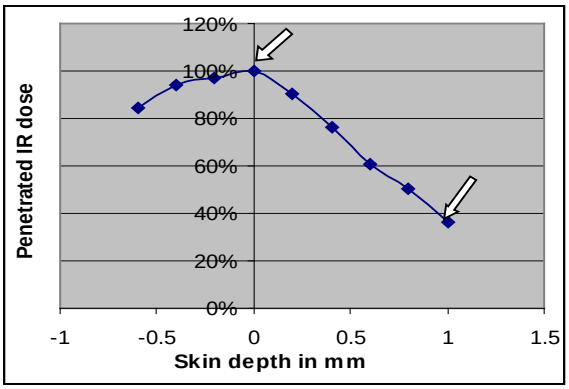


Figure 2. Distribution of radiation doses on dorsal skin surface of rat. The bottom square was the unirradiated control region.

The electron beam was generated by the Van de Graaff accelerator in the Nelson Institute. The LET of the electron beam was 0.34keV/μ, and the dose rate was 7.4 Gy/min. The penetration depth of the electron radiation was measured and is shown in Figure 3 below. At the chosen time of irradiation, the hair follicles were in their resting phase, thus their roots were located about 1mm deep in the skin. As the arrows in the figure point out, the deepest part of the hair follicles (at 1.0 mm) received approximately 40% of the dose to the surface of the epidermis. The dermis and hypodermis (between 1 mm and about 1.7 mmin depth) received a diminished dose generally less than 40% of that to the surface epidermis.

Figure 3. Penetration-dose curve of the electron radiation used in this study. Arrows point to the skin surface and the approximate location of hair follicle roots (1.0 mm), respectively.



Skin Tissue Collection and RNA Extraction

At 1 day and 7 days post-irradiation, the radiation exposed and non-exposed dorsal skin patches were excised from each rat and were pooled according to radiation dose and vitamin A supplement, i.e., 12 pooled samples of 4 rats each included 6 on the control diet receiving radiation doses of 0, 9, or 18 Gy irradiation and sampled at 2 time points, and 6 on the vitamin A diet with another 3 doses and 2 time points.

Central portions of each irradiated skin, about 0.7 cm x 0.7 cm in size, were excised, cut into small pieces and immersed into ice-cold Trizol (Gibco BRL, NY) reagent, then homogenized by using an electronic homogenizer (BioSpec Product Inc., OK) at top speed (20,000 rpm) for about 2 minutes. The pooled tissues were kept at -80°C until further processing.

Total RNA was extracted following the TriZol protocol. RNA quality was checked by spectrophotometer and denaturing agarose gels. Then the RNA was reverse transcribed to prepare double strand cDNA using a dT_{24} primer adjacent to the T7 RNA polymerase promoter region [primer sequence: 5'- GGC CAG TGA ATT GTA ATA CGA CTC ACT ATA GGG AGG CGG-$(dT)_{24}$ -3'] (from Affymetrix Inc., CA), and using the kit SuperScript™ Choice System for cDNA Synthesis (Gibco BRL, NY). The double-stranded cDNAs then were transcribed into biotin-labeled cRNAs and at the same time amplified through *in vitro* transcription (Enzo Diagnostics, NY), as suggested by the gene chip manufacturer. The resulted cRNAs were purified from un-incorporated nucleotides by a RNA Clean-up kit (RNeasy Mini Kit, Qiagen, CA) following manufacturer's instructions.

The purified, labeled cRNAs were fragmented enzymatically and a hybridization mix was generated as recommended by gene chip manufacture.

Gene Chip Hybridization

The cRNAs were differentially displayed by means of Affymetrix (Santa Clara, CA) rat genome gene chips (RAT U34-A) on Affymetrix chip systems in Dr. Eric Tang's lab and in Dr. Mike Blumenberg's lab in the NYU Medical Center in Manhattan. Each chip contains 8740 sets of specific probes for rat genes, including 3418 sequence tags plus about 5322 known genes representing every known functional category. With the gene chip, expression of all 8740 genes and gene groups can be examined in a single hybridization.

Hybridization of the gene chips were according to Affymetrix's guidelines. Briefly, aliquots of each sample (10µg cRNA in 200µl hybridization mix) were hybridized overnight to a rat oligonucleotide array. After hybridization, each array was washed, stained with streptavidin phycoerythrin (Molecular Probes), washed again, hybridized with biotin labeled antistreptavidin phycoerythrin antibodies, restained with streptavidin phycoerythrin (Molecular Probes), and scanned (Hewlett-Packard, GeneArray scanner G2500A).

Data Analysis

Data generated from the gene chip hybridizations by the GENECHIP 3.3 software (Affymetrix, CA) were input into Microsoft Excel for further analysis. Here, the expression value for each gene was determined by calculating the average of differences of the intensity from perfect matched probes minus the intensity from mismatched probe pairs for the genes. For data analysis, all the arrays were normalized to the total intensity of fluorescence, and all the genes whose expression levels were negatives (as a result of the subtraction of the signals of mis-matched probes from perfect matched probes) were assigned to value 1 (in some cases to 500, as will be specified later). Fluorescent intensity at 1000 was used as a cut-off to distinguish "expressed" and "not-expressed) genes, while the gene chip software "present" or "absent" calls were overriding for fluorescent intensities above 1000 with an absent call and for intensities below with a present call. About 50% of the genes have expression values above 1000, almost the same as the percentage of genes (though not exactly the same genes).

To find the functions of each gene, the GenBank number for the gene listed by Affymetrix was used in NCBI (National Center for Biotechnology Information) PubMed search for related articles describing this gene. For unknown genes, this was done by first obtaining the sequence from NCBI's Nucleotide search, then using the sequence in NCBI's standard nucleotide-nucleotide BLAST search to find functions of similar sequences and any updated information that might be available. Genes were grouped into 15 function groups (include one for unknown genes) based on broad categories defined by Dr. Levine's group (45). The categories were deliberately broad so that almost all known genes could fit into one of them.

Northern Blot and Semi-Quantitative PCR

Northern blots were carried out according to the protocol by Sambrook (46). Briefly, denatured agarose gel of 15 µg of total RNA in each lane were blotted onto nylon membranes. To make probes for hybridization, a representative DNA fragment was

amplified through RT-PCR using SuperScript Reverse Transcriptase (Gibco BRL, NY) and regular PCR. PCR products were run on agarose gel, the bands were cut and purified by DNA Gel Extraction Kit (Qiagen, CA). Then the PCR fragments were radioactively labeled by α-P^{32} dCTP (NEN Life Science, MA) by means of random priming labeling (kit from Roche). Total RNAs were reverse transcribed to cDNA, and the cDNA were used as templates on the Light Cycler (Roche) for real time PCR. All samples were normalized according to their GAPDH contents.

Semi-quantitative PCR: Similar to regular RT-PCR except that the cycle number for each primer set was empirically determined, so that the reactions were stopped before the reaction reached plateaus. The optimal cycle number was affected by the primer sequence and the cDNA concentration. Usually, 22 to 32 cycles were required.

Retinol concentration in the skin was measured for rats that have or do not have the dietary RA supplement. The result (Figure XX) shows that by 7 days after dietary RA was begun, the retinol had accumulated in the skin but not in the blood. The skin concentration of retinol in rats on the RA was 4.5 fold higher than in control, while the plasma retinol levels were rather steady with or without additional RA.

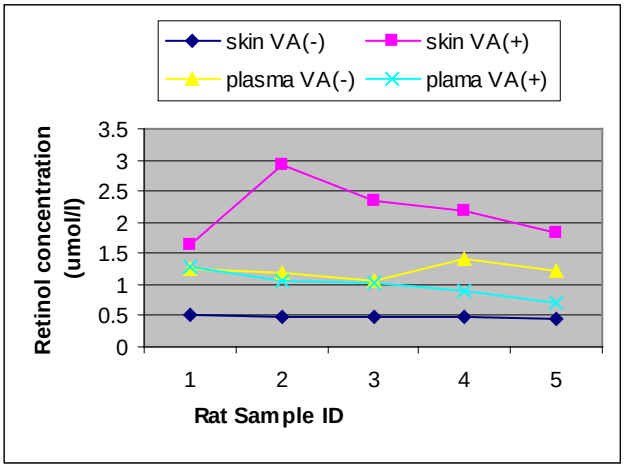


Figure 4. Retinol accumulation in rat skin and plasma after 7 days on retinyl acetate diet. As reference, plasma retinol levels are included for rats with or without RA supplement.

Results:

Human and rat skin are highly susceptible to the carcinogenic action of ionizing radiation. Both develop a mixture of basal and squamous cell carcinomas with the rat exhibiting in addition a small incidence of adnexal carcinomas and sarcomas (6). Prevailing evidence indicates that after repair of the initial DNA lesions and many rounds of cell division in viable survivors, a certain fraction of the initial molecular damage to DNA is converted to a variety of non-repairable point mutations, rearrangements, translocations, etc (47, 48). In the general multistage theory of carcinogenesis, cells progress to cancer through a series of heritable genetic alterations or mutations (49). The multistage theory of carcinogenesis was originally invoked to explain the temporal pattern of cancer incidence data in human populations (50-52).

The multistage approach has been reformulated in light of the experimental results in rat skin (53). Following a single dose of ionizing radiation to rat skin, epithelial and connective tissue tumors appear in a consistent time pattern that is remarkable and is the basis for constructing time-independent dose-response relationships as follows:

$$Y(D,t) = f(D)g(t) \quad (1)$$

where $Y(D,t)$ is the overall cancer yield in tumors per animal, D is the radiation dose and t is the elapsed time.

The functions $f(D)$ and $g(t)$ require specific analytic forms and for this the dual action hypotheses provides some guidance. A form of the dose-response function derived from dual action is:

$$f(D) / D = CL + BD \quad (2)$$

where L is linear energy transfer, D is radiation dose and C and B are empirical constants (54).

Experiments with irradiated rat skin suggest that $g(t)$ is a power function for a large fraction of the rat's lifespan. A frequently used power function form is:

$$g(t) = k(t - w)^n \quad (3)$$

where t is elapsed time since exposure and k, w and n are empirical constants, although ideally n should be an integer. Based on fitting equation 3 to cancer yield data in rat skin, n was estimated to be about 2 for epithelial cancers and fibromas, if w is negligible. This simple model fits the pattern of neoplastic responses in rat skin surprisingly well to the experimental results in rat skin surprisingly well, although further validation would be desirable.

To test the above formulation, rat skin has been exposed under identical conditions to 4 different types of particulate radiation with greatly different LET values: electron radiation, a neon ion beam, an argon ion beam or a ^{56}Fe beam. The LET of these radiations was 0.34 keV/µ, 25 keV/µ, 125 keV/µ and 150 keV/µ respectively. The neon ion and argon ion beams were generated at the Bevalac at the Lawrence Radiation Laboratory (55) and the ^{56}Fe beam was provided by the Alternating Gradient Synchrotron (AGS) at Brookhaven

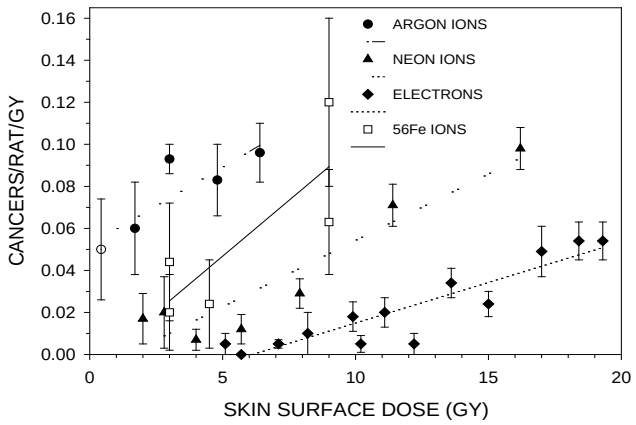


Figure 1. Cancer yield per unit dose as a function of surface dose at 1 year after irradiation for rat skin irradiated at 28 d of age with monoenergetic electrons, neon ions, argon ions and ^{56}Fe ions. Regression lines are shown fitted to the each data set.

National Lab. The cancer yield per unit dose, shown in Figure 1, is in excellent agreement with the quadratic formula in equation 2, and the ^{56}Fe data fit reasonably well into the general pattern, albeit with larger error bars. The data suggest the effective LET of the ^{56}Fe beam was closer to that of neon than to argon. The entrance LET of the pure ^{56}Fe ion beam was calculated to be 150 keV/μ, i.e. greater than argon, but the beam was contaminated by knock-on products with lower LETs, so an effective LET somewhere around 75 keV/μ is not unreasonable for ^{56}Fe based on the data in Figure 1.

Stem cells as targets of carcinogens

Evidence is accumulating that stem cells may be an important target for carcinogens, specifically because of the role of the c-*myc* oncogene in stem cell proliferation (56, 57). The 2 stage model of carcinogenesis was recently altered to take into account the contribution of possible subpopulations (56, 58). Lavker et al. concluded that the target cells for chemical carcinogenesis in mice and humans are actually stem cells that reside in a 'bulge' on the side of each hair follicle (59). At least 6 or so putative marker genes of keratinocyte stem cells are currently available and will be utilized in Affymetrix microarrays and custom microarrays to establish the identity and follow the fate of these cells following irradiation and dietary retinoid, eg. keratin 15 (60), beta1 integrin, E-cadherin, beta catenin, and plakoglogin (61), CD 71 (62), alpha6bri10G7dim (63) and p63 (64). Down-regulation of the latter gene has been shown to be strongly blocked by a retinoid (65)

Retinoids and differentiation of keratinocytes

Much work has focussed on the close connection between retinoids and keratin gene expression and differentiation in keratinocytes. A nuclear retinoid receptor has been shown to regulate keratin expression in human keratinocytes (66). Following retinoid treatment mRNA levels of keratins K5/K14, K6/K16 are reduced relative to actin mRNA, while K19 mRNA levels are increased (67). Topical retinoids tend to reduce the number of tumors in initiation-promotion experiments in mouse skin (68). Retinoids can alter oncogenes directly, e.g., c-*myc* oncogene expression can be reduced by a retinoid (69), while v-erbA may act in neoplasia by interfering with the retinoid receptor. In human lung cells retinoic acid increases the p53 mRNA while decreasing the mRNA of keratin K18 (70). Retinoids have been recognized as chemopreventive agents, although no special focus has been placed on radiation-induced cancers (71).

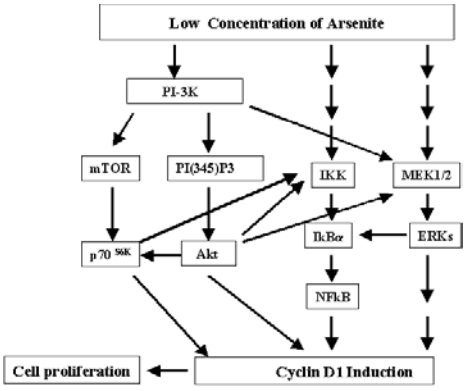


Figure 5. Schematic of Cyclin D1 Activation Pathways.

Cyclin D1 induction pathways

An upregulation of cyclin D1 expression plays an essential role in cancer development. Growing evidence suggests that the signal transduction pathways leading to the activation of ERKs, NFκB, and PI-3K-Akt-p70S6k are involved in the induction of cyclin D1 and cell proliferation caused by different chemical carcinogens in various cell culture models. Thus, we propose here to investigate the role of ERKs, NFκB, and PI-3K-Akt-p70S6k in the

induction of cyclin D1 and cell proliferation as cells respond to 56Fe ion radiation.

The role of p53 in carcinogenesis

In response to DNA damage, the p53 tumor suppressor gene is in part responsible for cell cycle arrest and, if DNA damage is severe, for the induction of apoptosis (72-75). Cells lacking in p53 exhibit partial deficiencies in DNA nucleotide excision repair (NER) repair of UV radiation induced DNA damage, whereas transcription coupled repair (TCR) is not dependent on p53 (76). Serine 15 phosphorylation of p53 is one of several modifications associated with increased transcription efficiency, decreased affinity for MDM2 (murine double minute-2), and increased nuclear localization of p53 In fact Zhang and Xiong identified a nuclear export signal in the amino terminus of p53 (residues 11 to 27) (77).

When WI38 normal human fibroblasts were irradiated with ionizing radiation (6 Gy), p53 and p21 protein concentrations were increased after 4h, as expected. Ford and Hanawalt (78) have shown that wild-type p53 is required for global genomic repair (GGR), and Smith et al. (76) has shown that human cell lacking functional p53 are partially deficient in the GGR but not in the TCR pathway of NER. With specific regard to repair of UV induced DNA damage, the loss of the downstream p53 effector, gadd45, is considerably more detrimental to cells than is the loss of p21 (76). In fact, human cells deficient in p53-mediated p21 expression still exhibit normal NER of UV damage (79). In summary p53 signaling pathways are manifold but a close tie to the initial cell cycle blockage following a DNA damaging event has strong experimental support.

Direct measurements of *in vivo* gene expression patterns in epidermis indicate ionizing radiation alters the expression of about 180 genes while a dietary retinoid alters nearly 220 genes. However only about 30 genes are altered by the radiation and the retinoid in reverse directions including; a) keratin 14 – a marker of basal keratinocyte proliferation, b) calcium binding protein, c) ornithine decarboxylase (ODC) and d) involucrin. The retinoid-induced alterations in the latter genes have the effect of reverting the cells to a more differentiated, i.e. a less proliferative state. That this reversal also prevents about 50% of the cancers following a dose of radiation points to a possible way that the retinoid may exert its preventive action, i.e. by prolonging the radiation-induced cell cycle arrest of the pluripotent stem cell population; an attractive idea because stem cells are likely targets for cancer induction (62). Putative markers of skin stem cells exist and suggest that stem cells are probably a very small subpopulation normally, although they can rapidly expand when stimulated. The pluripotent stem cells will be sought through the use of marker genes and unique gene expression patterns following exposure to ^{56}Fe radiation and/or a dietary retinoid in comparison to other cutaneous keratinocytes (80). Rat keratinocyte cell lines will be studied to confirm and extend the *in vivo* data and to look for possible consensus sequences in the promoter region of retinoid-responsive genes as an approach to a better understanding of how the retinoid is counteracting the gene expression response to the radiation.

Inhibition of skin cancer by dietary retinoid

The original motivation for studying the cancer inhibiting properties of vitamin A acetate was based on results obtained with low LET electron irradiation of rat skin. These experiments showed, as seen in Figure 3, that a dietary retinoid beginning 2 weeks prior to irradiation and continuing for life, prevented nearly 90% of skin cancers of all types (81). Even an exposure period beginning 2 weeks prior to irradiation and continuing for 1 week after irradiation was about 50% effective, while beginning 1 week after irradiation and continuing for life was also about 50% effective.

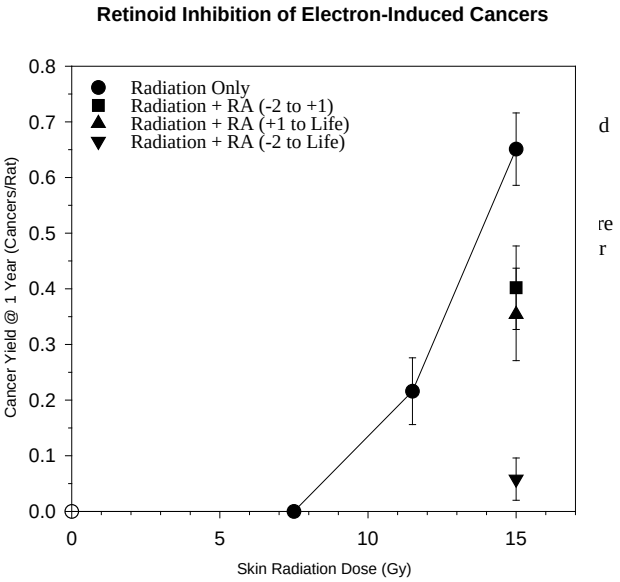


Figure 6. The Effect of dietary retinoid on skin tumor induction in rats exposed to electron radiation.

If the stem cell hypothesis is correct, a reduction in cancer incidence should be reflected in an equivalent reduction in the number or proliferative activity of stem cells responding to the radiation. What happens to these stem cells? They probably differentiate and keratinize (82). Rat skin consists of about 10^7 epithelial cells per cm^2, but the maximum number of malignant neoplasms seen, even with the most powerful carcinogens available, is rarely more than about 1-2 per cm^2. This implies that the epithelial cells are not very likely to undergo carcinogenic alterations, and/or that the population at risk is a small fraction of the total population. Certainly stem cells ought to be considered a prime target population for neoplastic progression (59, 83).

Prior to preliminary studies in the current series, little was known about whether the cancer preventive action of vitamin A observed with low LET radiation would apply to cancers induced by high LET radiation, such as, the ^{56}Fe ion beam at the AGS in Brookhaven Laboratory. Preliminary experiments with dietary vitamin A acetate in conjunction with exposure to the ^{56}Fe ion beam have shown conclusively that a preventive effect is present for both malignant and benign tumors, although at a lesser magnitude than for electron radiation.

Gene expression microarray analysis of rat skin

Gene expression microarray analyses have been carried out on the skin of rats exposed to 0 or 9 Gy of electron radiation and fed or not a vitamin A supplemented diet. Based on 5-fold differences 9 Gy relative to control showed 188 altered genes; vitamin A treatment relative to control showed 231 altered genes, while 9 Gy relative to 9 Gy plus vitamin A showed 58 altered genes. One approach to identifying genes of interest is to find genes that are reversely regulated by vitamin A and radiation. Among about 30 genes with opposing responses, the ones with the greatest reversal were: keratin 14 (K14), ornithine decarboxylase (ODC), an MRP-14 (intracellular calcium-binding protein gene), and an EST rc_AA957003 which has sequence homology to intracellular calcium-binding protein

MRP 8. The alterations of each of these genes suggest the retinoid is blocking radiation's pro-poliferative effects.

Gene expression responses were segregating into functional groups. Several groups failed to show any genes with significant changes after radiation with or without vitamin A, including apoptosis genes and oncogene/tumor suppressor genes. The 100 most highly expressed genes in controls were assigned to functional categories as a control for how genes might be distributed if expression levels were affected only in proportion to the number of genes in the category, i.e. randomly (Figure 7).

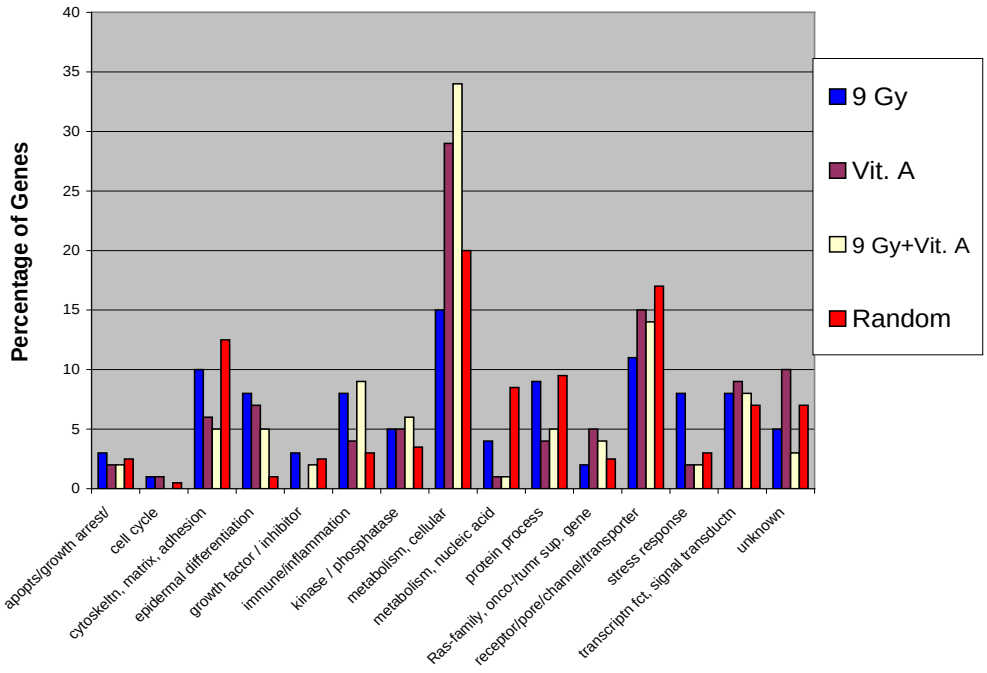


Figure 7. Comparison of the functional groups that were affected by radiation and vitamin A. The 100 most altered genes for 9 Gy alone or vitamin A alone or 9 Gy + Vit. A versus controls were assigned to functional groups; the distribution of the 100 most highly expressed genes in controls are included (random) for comparison purposes.

The data in Figure 7 show that the most frequently affected genes by 9 Gy or vitamin A alone were cellular metabolism and receptor/transporter genes. Epidermal differentiation genes were much more frequently affected in comparison to random as seen in the 4th group of columns. Northern blots showed a similar trend of expression as the gene chips.

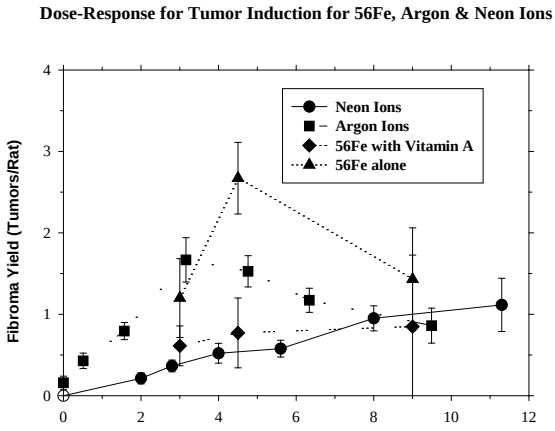


Figure 8. Dose response relationships for all tumors at 1 year after exposure to the various beams of radiation as indicated. The effect of the vitamin A on tumor induction was equivalent to lowering the 56Fe LET to that of neon ions, i.e. to about 26 keV/μ.

Inhibition of 56Fe induced skin tumors by dietary retinoid (Figure 8)

An understanding of the differing biological effects of high and low-LET radiation is important for predicting the cancer risk of radiation exposure. High-LET radiation causes

greater initial DNA damage, unrepaired DNA strand breaks, and cell death than low-LET (3, 4, 84). The different features of high and low-LET radiation correlate with differences in molecular events associated with progression of rat skin tumors (5). The all tumor results of the 1.1 geV/nucleon ^{56}Fe exposures along with comparable results from previous experiments with argon and neon ions, are plotted in Figure 5. The data show that single doses of ^{56}Fe ions are somewhat more effective than argon in producing tumors at 4.5 Gy but are about equally effective at 3.0 Gy and 9.0 Gy. A strong tendency is apparent for the tumors to be inhibited by the presence of 250 ppm retinyl acetate in the lab chow.

Signal transduction of the cyclin D1 pathway in epidermal keratinocytes

In the proposed studies, we will first use human skin keratinocytes (HaCat cell line). HaCat cell line is selected because this cell line is a spontaneously immortalized, but not tumorigenic cell line (85), and has been shown responsive to metal exposure similarly to primary cultures of human keratinocytes (86, 87). We have tested this cell line in our Laboratory and found that HaCat can't grow in soft agar. However, it can grow in soft agar in the presence of tumor promoter EGF (10 ng/ml) (data not shown). To verify the results found from HaCat cell line in the proposed studies, we will test all major findings from HaCat cells in primary cultured human keratinocytes. We have already performed several preliminary experiments related to the proposed work and obtained the following data to justify the specific aims of the current grant proposal.

As a model for work proposed here, our laboratory has conducted research on arsenite-induced signal transduction in mouse epidermal Cl 41 keratinocytes and have investigated the role of the PI-3K pathway in TPA- or EGF-induced AP-1 activation and cell transformation (88, 89). We have also established stable transfectants with dominant negative mutants of PI-3K, ERK2, PKC, Akt, IKK, IκBα, etc in different cell lines and have studied the role of these kinases in vanadate- or UV-induced signal transduction pathways, as well as, biological functions (88 , 89-98).

An up-regulation of cyclin D1 expression plays an essential role in proliferative activity of cancer cells. Growing evidence also suggests that the signal transduction pathways leading to the activation of ERKs, NFκB, and PI-3K-Akt-p70S6k are involved in the induction of cyclin D1. Thus, we propose here to investigate the role of ERKs, NFκB, and PI-3K-Akt-p70S6k in the induction of cyclin D1 and cell proliferation as cells respond to retinyl acetate and ^{56}Fe ion radiation. The proposed cyclin D1 signal transduction studies are a logical continuation of our previous work.

Induction of PI-3 kinase in human keratinocytes

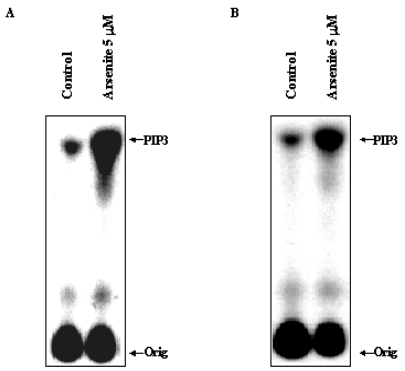


Figure 9. Induction of PI-3K activities by arsenite in primary human keratinocytes (A) and in the human keratinocyte cell line, HaCat (B).

As a preliminary we exposed human keratinocytes to arsenite to test for PI-3K activation. Human keratinocytes were cultured in monolayers in 100-mm plates. The cells were starved in 0.1% FBS DMEM medium for 48 h at 37°C. Arsenite (5 μM) was then added to cell cultures, and cells were harvested 15 min after the exposure. PI-3K activities were measured (88, 89). Exposure of cells to arsenite resulted in marked activation of PI-3K in both primary human keratinocytes and a HaCat cell line. These data (Fig. 9) suggest that PI-

3K is involved in the response of human keratinocytes to arsenite. To investigate the possible induction of cyclin D1, we exposed human keratinocyte HaCat to arsenite (5 μM) for 18 h. Cells were harvested and extracted with SDS-sample buffer. The cell extracts were separated on polyacrylamide-SDS gels, transferred, and probed with rabbit antibody against cyclin D1. The results showed that arsenite exposure markedly induced cyclin D1 after an 18-h exposure. We expect that ^{56}Fe ions will be an equally potent inducer of keratinocyte proliferation with a similar induction of cyclin D1.

UV-induced NFκB activation in mouse epidermal Cl41 cells

To determine whether arsenite treatment is able to enhance UV-induced activation of signaling, we co-incubated arsenite with mouse epidermal Cl41 NFκB-luciferase stable transfectant, Cl41 NFκB mass1, after cells were exposed to UVB radiation. The results showed that arsenite treatment led to a synergistic effect on UVB-induced NFκB activation in Cl41 cells (Fig. 10). These results revealed that the synergistic effect of arsenite on UV-induced NFκB activation is correlated with its co-carcinogenic effect on UV-induced skin tumor induction, suggesting that arsenite may have synergistic effects on UV radiation-induced NFκB activation, as well as, other signal transduction pathways in human keratinocytes.

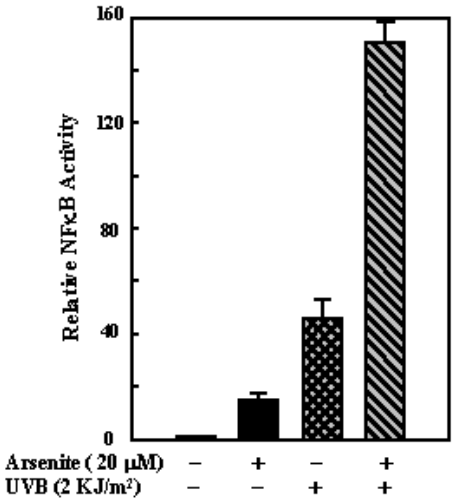


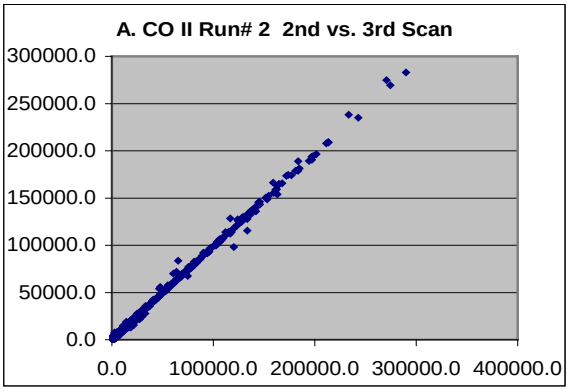
FIG. 10. An effect of arsenite on UVB-induced NFκB activation. Cl41 cells stably transfected with a NFκB-luciferase reporter (Cl 41 NFκB mass1) were seeded into each well of 96-well plates, and cultured in 5% FBS MEM at 37 °C. The medium was replaced with 0.1% FBS MEM after the cell density reached 80-90%. Twelve hours later, cells were treated with UVB radiation (2 KJ/m2) and/or arsenite (20 μM) for induction of NFκB. The cells were extracted with lysis buffer, and luciferase activity was measured at 24 hr after exposure using the Promega luciferase assay. The results are expressed as NFκB activity relative to control medium (Relative NFκB activity). Each bar indicates the mean and standard deviation of four repeat wells.

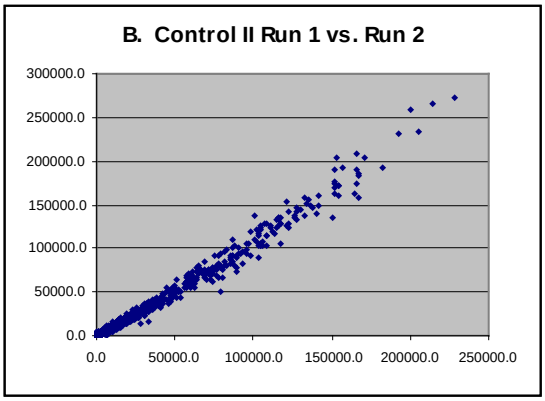
For samples collected at 1 day post-irradiation, five gene chip hybridizations have been carried out for control diet rat samples with 0, 9 Gy and 18 Gy radiation (designated as control, 9Gy and 18Gy, respectively), plus vitamin A treated samples with 0 and 9 Gy radiation (designated as vitamin only and vitamin + 9Gy). Another five gene chip hybridizations were carried out at 7 days post-irradiationproviding: two hybridizations of the control control where the probes were prepared by identical but separate procedures as

a check on reproducibility, plus 9Gy, vitamin only, vitamin + 9Gy for evaluation of the persistence of the 1 day responses.

The expression intensity of each gene was normalized across different chips by using the total average fluorescent intensity of all 10 chips. The resulting expression signals ranged from 0 to 278,710 (for EST228160), with an average of 4813. When signals above 1000 were considered significant expression, then for 1 day data 4402 of the 8740 (50.4%) genes were positive in at least one of the 5 samples, and the remaining 4338 genes (49.6%) were considered as not significantly expressed across all the 5 chips. For 7 day data, the positive gene ratio was 50.85% (4444 genes) while the not-expressing percentage was 49. 15%. Since genes with lower expression levels tend to have larger variations, the following analysis used a cut-off of ≥1000 intensity to distinguish positively from negatively expressed genes. The Affymetrix software called about 50% of the genes as "Absent". This ratio is very close to the non-significant ratio generated by the cutoff at 1000, although not always the same "Absent" genes were cutoff (80% overlap).

Reproducibility of the gene chip technology was checked by the hybridization of the same control RNA, by using separately prepared cRNA probes and also by comparing multiple scans of the same chips. Figure 11A is an example of duplicate scanning of the same gene chip. Figure 11B shows a scatterplot of two hybridizations to the same source mRNA but with cRNA probes prepared separately. The almost perfectly matched scatterplots in panel A and B demonstrate the reproducibility of the experimental approach. As a comparison, Figure 11C shows the expression patterns of control versus a 9 Gy sample, displaying a number of genes that were significantly regulated by the radiation treatment.





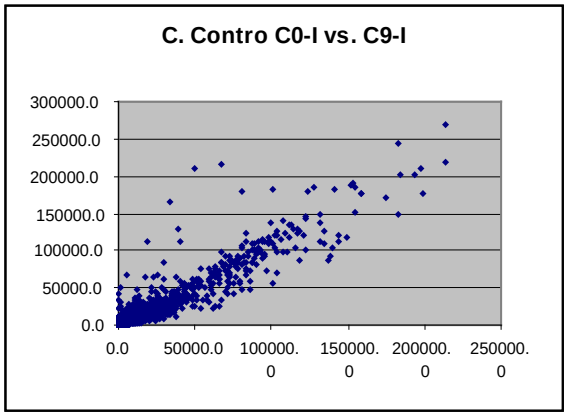


Figure 11. Scatter plots showing reproducibility and variability of the experiments. A: Different scans of the same 7 day control #2 (Run 2 of control-II) chip. B: Results from individual probe preparations of the same 7 day control RNA (Run 1 vs. Run 2 of control-II). C: Real difference in 1 day control control-I versus a 9Gy irradiated sample 9Gy-I.

Radiation Effects on Gene Expression of Rat Skin at One-Day Post-irradiation

The expression levels of each irradiated sample were compared with those of control control. Transcripts were selected if they showed at least a 5-fold change, and at least have intensity greater than >1000 in one sample, and the stronger expressed gene had to be "present" by Affymetrix call. By these criteria, 239 and 110 regulated genes by 9 and 18 Gy irradiation were identified, respectively, accounting for about 2.7% and 1.3% of the genes represented in gene chip used. When combined, 284 genes were regulated by either dose of the electron radiation at the time point of 1day post-irradiation.

Table 1 lists the 40 most greatly affected genes at 1 day after 9 or 18 Gy electron radiation. Since lower expression values have higher variability based on the limitations of chip technology, intensities <500 were adjusted to equal 500 in calculating fold changes (i.e., 9Gy/control or 18Gy/control) to avoid generating artificially high or misleading fold ratios.

Genes Involved In Stress Response And Inflammation

Gene expression data reflected the fact that the radiation caused a stress response and inflammation of the skin tissue at 1day post-irradiation. Following irradiation stress genes, like cytochrome P450 2F4 (AF017393), and 2 versions of myelencephalon specific protease (AF016269), showed 2.4, 8 and 27-fold increases, respectively; while inflammation-related genes like beta defensin-1 (AF093536), beta defensin-2 (AF068861), and intracellular calcium-binding protein 14 (MRP14, L18948) showed induction from almost no-expression. Together with the strong induction of another calcium-binding inflammatory gene (EST rc_AA957003) which is similar to the human intracellular calcium-binding gene 8, the data suggest the electron radiation-evoked skin inflammation was accompanied by activation of the intracellular calcium-pathway, a very important pathway for epidermal differentiation and proliferation.

Genes involved in oxidation response and damage repair

The oxidation damage of the radiation to the rat skin is reflected by the up-regulation of several oxidation-relate transcripts like urate oxidase (uricase, J03959 and X13098), glutathione-dependent prostaglandin synthase (AF021882) and dithiolethione-inducible gene-1 (DIG-1, U66322). However, other reported oxidation-induced genes, such as, glutathione peroxidase and superoxide dismutase, were not altered at 1 day after irradiation in the rat skin tissue, possibly due to the difference in tissue type and/or different time intervals after irradiation.

The dose of electron radiation used in this study was sufficient to cause substantial DNA and protein damage. As a result it was expected that increased DNA repair activities might be activated within 1 day after the electron radiation. However, transcripts of DNA repair genes were not altered significantly at this time point. This could be partially explained by repair genes always being active and that a cutoff at 5 fold would likely miss smaller alterations. Another possibility is that damaged cells could regulate their repair activity by modifying repair enzymes rather than regulating mRNA levels. Such results have been noted in UVB irradiated human skin keratinocytes (Li D at al., 2001) regarding the repair genes.

Genes involved in apoptosis and cell cycle regulation:

In response to genotoxic radiation damage, cells may initiate apoptosis or growth arrest, proliferation or differentiation, depending on cell types and damage types. Here in this data set no obvious change of gene expression was found regarding apoptosis,

Table 1 **Genes That Were Most Greatly Altered at 1 day Post-eRad**

	9Gy/control	18Gy/control	Description
1	0.2	0.1	J03179 D-binding protein
2	0.2	0.2	U22424 11-beta-hydroxylsteroid dehydrogenase type 2
3	0.1	0.2	rc_AA866452, ~ to alpha-actin
4	0.3	0.2	U50185 protein phosphatase 1 myosin binding subunit
5	0.1	0.2	AB011532 MEGF6, with multiple EGF
6	4.9	8.7	AF017393 cytochrome P4502F4 (CYP4502F4)
7	16.9	9.4	AF104399 melanocyte-specific gene 1 protein (msg1)
8	27.3	15.3	AF016269 myelencephalon specific protease
9	12.2	7.1	rc_AI010453, a serine/cysteine proteinase inhibitor
10	8.6	2.4	D00753 contrapsin-like protease inhibitor related protein
11	4.4	3.8	M32397 prostatic acid phosphatase (rPAP)
12	3.0	4.9	X64563 plasminogen activator inhibitor 2 type A (PAI2A)
13	13.1	8.9	AF093536 beta defensin-1 (BD-1)
14	9.8	9.6	AF068861 beta defensin-2
15	8.0	13.0	J03959 uricase 3 end
16	5.5	9.6	X13098c urate oxidase (uricase)
17	4.6	4.8	AF021882 glutathione-dependent prostaglandin D synthase
18	12.3	4.6	U66322 dithiolethione-inducible gene-1 (DIG-1)
19	45.5	58.2	L18948 intracellular calcium-binding protein (MRP14)
20	3.7	4.4	rc_AA957003, ~to intracellular calcium-binding gene 8

			MRP8)
21	11.8	4.6	D63774 RATRETK keratin 14, basal cell proliferation marker
22	4.2	4.4	L46593 small proline rich protein (spr)
23	5.8	5.4	rc_AA89191, =small proline-rich protein (spr)
24	68.9	33.3	clone rx02392, ~ to mouse keratin 1
25	15.8	8.5	rx01678, ~ to mouse keratin 1
26	20.0	12.2	AB003753 high sulfur protein B2E and B2F,
27	3.3	2.0	U49062 heat stable antigen CD24, differentiation marker of hair follicle
28	9.7	8.4	M57263 protein-glutamine gamma-glutamyltransferase
29	3.5	3.1	U63972 blue cone opsin-like pigment
30	4.1	5.1	AB014722 rSALT-1(806), SCC antigen, oncogene
31	22.1	18.3	clone rx04836, ~ to mouse SCC antigen, oncogene
32	19.9	2.9	S74257 2c9, c-fos transformation induced
33	8.4	8.1	rc_AA800305 EST189802, ~to mouse RAB5, RAS family
34	4.8	3.4	A30543cds p-Meta-1 CD44 surface protein
35	5.8	2.2	X82669completeSeq RT1.Au gene
36	3.1	2.7	M31076 transforming growth factor alpha (TGF-alpha)
37	9.2	4.8	D26393 type II hexokinase, exon1 and promoter
38	16.4	8.4	M13949 cellular retinol binding protein II (CRBPII)
39	24.4	11.8	clone rx05063, ~ to no known gene
40	12.1	11.6	rc_AA800665 EST190162, a membrane gene

implying apoptosis may not be initiated by 1 day post-irradiation, although this is not a proof because apoptosis may occur through other unknown pathways. However, several protease inhibitors were increased at this time point. It is known that certain kinds of the protease inhibitors promote apoptosis. A certain consistency is apparent in that the expression of a very important gene in regulating apoptosis, p53, was not altered significantly at 1 day.

Most cell cycle-related transcripts were not significantly regulated following the radiation, except that cyclin G was down-regulated by 5.7 fold, this cyclin may have a closer association with growth stimuli than with the cell cycle per se (99). With only one cyclin gene (out of 16) significantly altered, it is unlikely that the radiation had arrested the epithelial cells at 1 day.

Genes involved in epidermal proliferation and differentiation

Expression of the proliferation marker, PCNA (proliferating cell nuclear antigen), did not change significantly after radiation. However, the keratin 14 (D63774) gene showed a 12 fold induction by 9 Gy. This gene is a marker of skin basal cells which are the only proliferating cells in the rat epidermis. The strong induction of this gene suggests the radiation may trigger a mechanism for the basal cells to proliferate prior to regenerative replacement of the damaged epithelial cells. As a result of such compensatory growth, the skin eventually will became hyperplastic and increased its thickness in the following weeks. Consistently, two transcripts for squamous cell carcinoma (SCC) antigen, EST rx04836 and rSALT-1 (AB014722), were strongly up-regulated by the electron radiation. At this time, we don't exactly know the function of

this antigen, but since it is a marker of epidermal cancer, most likely it is related to epidermal proliferation or invasion.

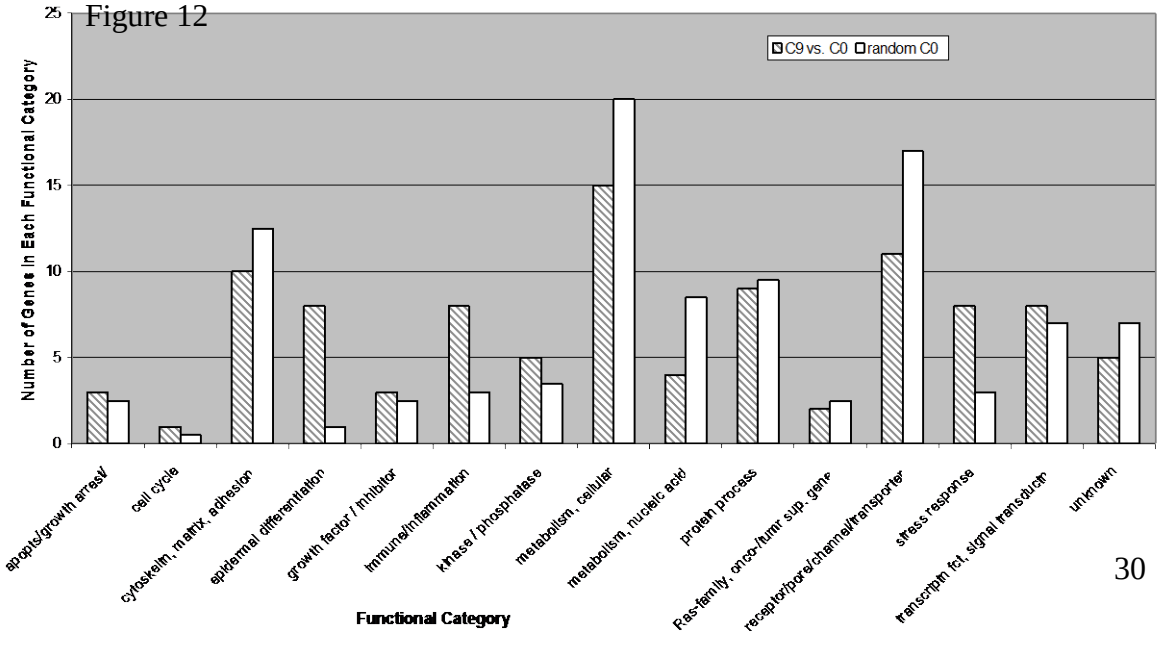
Unexpectedly, the radiation also increased several pro-differentiation transcripts, such as keratin type 1 (X54806) and it's 2 homologues (rx_02392 and rx_01678), heat stable antigen CD24 (U49062) which is a differentiation marker of hair follicles, and profilaggrin (M21759) which is an epidermal differentiation marker, etc. The seemingly paradoxical mixture of pro-differentiation and pro-proliferation signals from the data can be explained by the theory that radiation has caused proliferation in one population and at the same time promoted a premature differentiation in a different population. This hypothesis is similar to the immuno-histological observation that mouse skin epithelium developed hyperplasia and at the same time displayed a pattern of disturbed terminal differentiation after gamma-irradiation (100). A more recent study by Sivan (101) on human skin also demonstrated ionizing radiation caused hyperproliferation and abnormal differentiation.

Gene groups that were most greatly affected by electron radiation

To identify gene categories that were most greatly affected by the electron radiation treatment, the top 100 most-affected genes were picked out and assigned into functional groups based on biological information from GenBank. The numbers of genes in each functional group are plotted in Figure 12. For reference purposes, random is an artificial grouping based on a random selection of 200 genes from the expressed gene list on the untreated control chip. Such chart visually reflects the changes in the total number of affected genes in the various categories, no matter if the changes are in the direction of up- or down- regulation. Some genes may have several known functions, but only the best-known functions were used to make the figure.

By comparing 9 Gy to the randomly selected genes, we see that radiation affected more genes involved in epidermal differentiation, inflammation, nucleic acid metabolism and the stress response than would be expected in control skin. On the other hand, 9 Gy didn't significantly affect the number of genes in the categories of apoptosis, cell cycle nor protein processes at either 1 day or 7 days postirradiation. Efforts were made to distinguish between genes in functional groups and their expression patterns

Figure 12



(induction/repression). In many categories, correlations were not evident, perhaps due to the complexity of the categories. However, in several groups, most genes have the same trend of regulation, possibly indicating that the biological process is globally regulated. In the 9-Gy group, global induction can be seen in inflammation/immune and stress response, while global repression is seen in the receptor category.

Functional grouping of genes that were affected by 9-Gy electron irradiation at 1 day post-electron radiation. Top 100 most greatly affected genes were assigned into one of the above function groups, then the total numbers of genes in each functional group were plotted (striped bar). As reference, 100 randomly selected genes from the entire "expressed" gene list from untreated skin were assigned into functional groups (repeated twice, for an average) and plotted (open bar).

Radiation Effects on Gene Expression of Rat Skin at 7 days Post-irradiation

The expression levels of each irradiated sample were compared with those of control, control. Only 28 genes showed more than 5 fold regulation and at the same time had at least an expression intensity greater than 1000 in one sample and had "present" call by the Affymetrix software for the stronger expressed gene. When the fold change was extended to 3.0, 63 genes (see Appendix for details) showed regulation. Among these 63 genes, 37 transcripts that showed the greatest alteration were selected and listed in Table 2.

The pattern of altered transcripts at 7 days after electron radiation was not quite the same as at 1 day. Not only was the total number of the altered genes reduced with the lapse of time after the electron radiation, but also different genes were affected at the two time points, although there was also some overlapping. For example, in addition to the inhibition of an apoptosis-suppressing gene, namelyBcl-2 related ovarian death gene (AF065433), three other pro-apoptosis genes Bcl-xbeta (U72349), voltage dependent anion channel (AF048828) and sip9 (AF012887) were increased significantly at 7 days post-electron radiation. These facts suggest apoptosis may be at least partly underway at 7 day after the 9 Gy electron radiation, although at 1day post-electron radiation, no alterations of apoptosis-related gene expression could be seen.

In comparison to skin samples at 1 day post electron radiation, transcripts reflecting epidermal differentiation were altered to a much lesser extent at 7 days. One transcript, profilaggrin (M21759), whose gene product filaggrin is a major component of the cornified envelope, showed a persistent increase from 1 day to 7 days post the electron radiation. Inflammation related genes were also increased, 3 transcripts from the major histocompatibility complex (MHC) cluster displayed electron radiation enhancement at 7 days post-electron radiation. The MHC gene products are immune and inflammation-related, the up-regulation of them could demonstrate part of a long-lasting stress response elicited by radiation. While at 1 day post-irradiation, different

Table 2 List Of Genes That Were Most Greatly Altered At 7 days After 9 Gy electron radiation

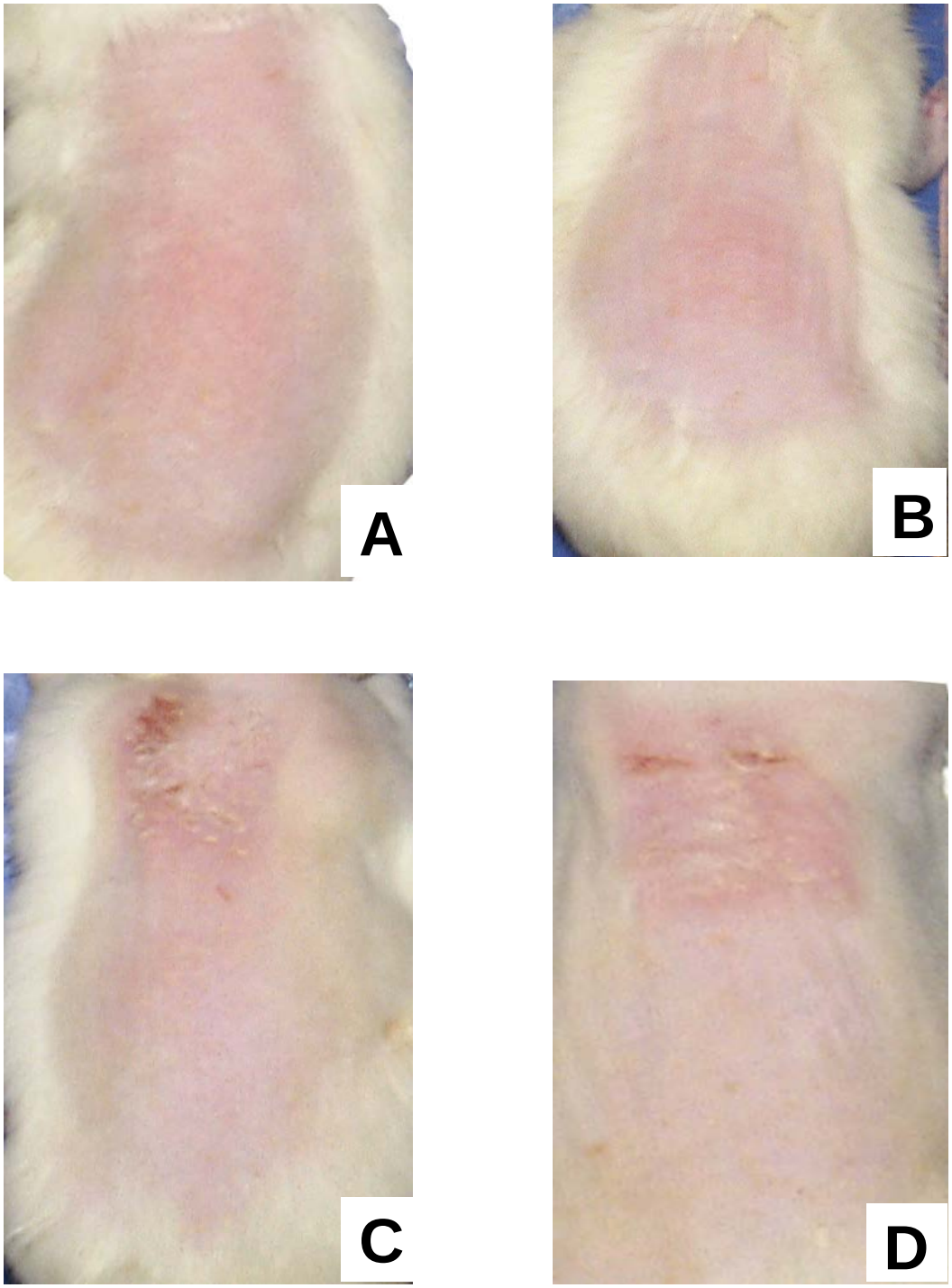
9Gy/control fold change	Gene Descriptions

1	**0.2**	AF065433 Bcl-2 related ovarian death gene, anti-apoptotic
2	**0.2**	AF091576 isolate HFL-VN1 olfactory receptor
3	**0.2**	rc_AA859633, ~ to no known gene
4	**0.2**	cDNA clone rx00875, ~ to no known gene
5	**0.3**	S78556 75 kda glucose regulated protein (grp75)
6	**0.3**	D10709 Isk protein, 3 terminus of exon 2
7	**0.3**	S66184 lysyl oxidase (3 region)
8	**0.3**	L13619 insulin-induced growth-response protein (CL-6)
9	**0.3**	L13619 insulin-induced growth-response protein (CL-6)
10	**0.3**	X06942 A-raf, ~ to raf (a protein kinase)
11	**0.3**	rc_H31363 EST105334
12	**0.4**	AB010999 peptidylarginine deiminase type IV (PAD-R4)
13	**0.4**	U76714 cell adhesion regulator (CAR1)
14	**2.6**	X14254 MHC class II-associated invariant chain
15	**2.6**	K02815 MHC RT1-B region class II A-alpha glycoprotein
16	**2.7**	X14254 MHC class II-associated invariant chain
17	**3.7**	X14812 ventricular myosin light chain 1
18	**3.9**	L13606 skeletal muscle myosin heavy chain 3 end
19	**4.8**	X15939 beta cardiac myosin heavy chain
20	**5.2**	M11851 heart myosin light chain 2 (MLC2) 3 end
21	**6.9**	X07314 heart myosin light chain 2 (MLC2)
22	**8.3**	X81193 muscle LIM protein(MLP), myogenic differentiation
23	**13.5**	X07314 heart myosin light chain 2 (MLC2)
24	**2.8**	X73524 desmin, IF of muscle cells
25	**3.4**	M21759 profilaggrin
26	**4.1**	rc_AA800206 EST189703, ~ to mouse actinin alpha 2
27	**4.1**	S74257 2c9, c-fos transformation induced
28	**4**	cDNA clone rx04836, ~ to mouse SCC antigen 2
29	**4.4**	AF012887 sip9, stress-inducible, pro-apoptosis
30	**5**	AF048828 voltage dependent anion channel, pro-apoptosis
31	**5.1**	U72349 Bcl-xbeta, promotes apoptosis

32	**5.8**	U68417 branched chain aminotransferase precursor (mit.)
33	**6.5**	D83796 UDP-glucuronosyltransferase (UGT1)
34	**2.6**	M58308 histidase
35	**2.6**	AF016269 myelencephalon specific protease
36	**4.5**	AJ005641 brain serine protease bsp1
37	**3.4**	M13949 cellular retinol binding protein II (CRBPII)

Abbreviations: kc-keratinocytes; mit.- mitochondria; IF-intermediate filament.

Figure 13. Rat skin pictures at 7 days after 0, 9, or 18 Gy (highlighted) electron radiation. Two representative skins from dietary vitamin A treated rats are shown (A & B), as are two skins from rats on the control diet (C & D).



inflammation related transcripts were increased.

These data support the direct observation that irradiated rats develop erythema (dilated capillaries, a clinical sign of inflammation) and scales on their irradiated dorsal skins at 7 days, in comparison to the normal un-irradiated control skin. Figure 13 panel C and D show pictures of skins of rats on the control diet at 7 days after 9 or 18 Gy electron radiation. The radiation damage in the form of erythema and desquamation is obvious in these photos at 9 Gy and at 18 Gy, and the RA produced a significant inhibition to the extent that no damage is discernable.

Strikingly, there are eight myosin/myosin-related transcripts as listed in Table 2, accounting for about a third of all the up-regulated genes in the list. They stand out from the total of 36 "expressed" myosins on the gene chip. This shows a rather strong and statistically significant regulation in the dermal part of the skin, since myosins are mostly concentrated in fibroblasts (myoblasts) and muscle fibers in the dermis. To our knowledge this is the first report on myosin-induction in irradiated skin, while many other studies working on radiation effects on *in vitro* keratinocytes could not have seen such alterations, because the requisite cell types are absent. The presence of myosin induction suggests the involvement of the dermis in addition to epidermis in the electron radiation response in vivo. While the mechanism and biological meaning of this induction is not clear, it is possibly an early stage of fibrosis, especially at the higher radiation dose. However, vitamin A addition could hardly

At 7 days post-electron radiation, expression of 2c9 (S74257), which could be activated by c-fos transformation in rat fibroblasts, was increased. This gene has invasion and metastasis-promoting ability and was up-regulated since one day post-irradiation. An EST that's homologous to mouse squamous cell carcinoma (SCC) antigen, which is possibly proliferation and metastasis-related, was consistently up-regulated at 1 day and 7 days post-electron radiation. The retinyl acetate consistently inhibited the expression induction of both genes at the two time points studied. The data also show a persistent up-regulation of the expression of the cellular retinol binding protein II (CRBP II, M13949) after electron radiation, this induction may be a compensatory mechanism for the radiation-depletion of the epithelial retinoid pool.

In summary, 9 Gy electron radiation enhanced the expression of stress response, inflammation, epidermal proliferation, abnormal epidermal differentiation genes, and repressed repressed the expression nucleic acid metabolism and cellular metabolism genes in rat skin at 1 day post-irradiation; however, by 7 days, the response was milder with respect to induction of apoptosis and stronger with respect to induction of cell movement/migration related myosin transcripts. Meanwhile inflammation-related genes were induced both times, while cellular proliferation and differentiation genes were marginally increased from the control.

Gene Expression Profiling Of Radiation Effects:

The 284 genes that were altered by one of the radiation treatments were clustered into several expression patterns shown in Figure 14. The figure shows that most of the affected genes were up-regulated, especially at 1 day post-irradiation as opposed to 7 days. Some genes showed a radiation dose-response while others did not.

A summary of radiation actions on the rat skin as inferred from gene expression responses is depicted in Figure 8. Some generally-accepted responses like the electron

radiation-induction of AP-1 are presented in the chart. The figure depicts AP-1 activation followed by proliferation and abnormal differentiation, stress response, apoptosis (later response), and inflammation (persistent response) as well.

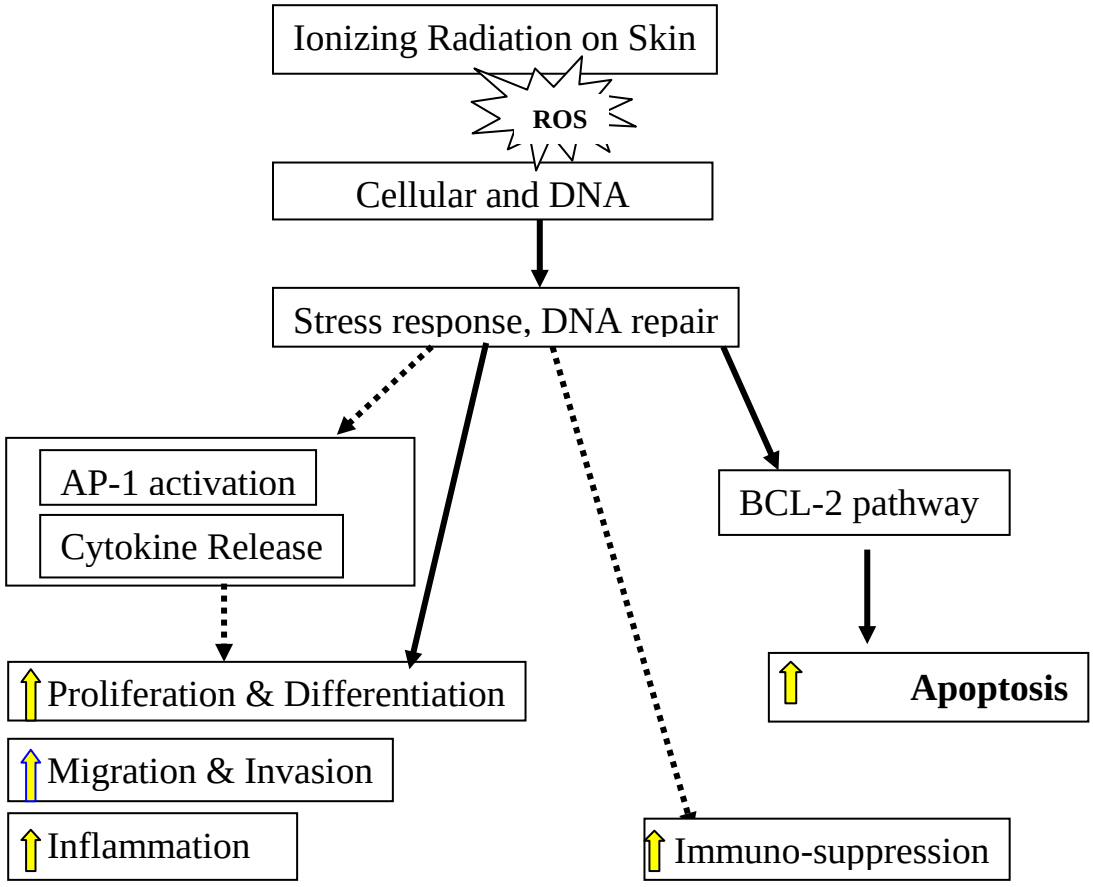


Figure 14. Schematic representation of ionizing radiation effects on the rat skin. Solid linking arrows show the responses identified in this study, and dotted linking arrow shows the responses reported by others.

Retinyl Acetate's Effects On The Rat Skin

Skin gene expression responses during dietary retinoid treatment were monitored for 29-days old and 35-days old rats (corresponding to the time of 1 day and 7 days post-electron radiation). Gene expression levels of vitamin A treated skin samples were compared with those of no-treatment controls. Three criteria were used to select the differentially expressed genes: (1) transcripts show at least a 2.5 fold change at one time point; (2) the intensity of the stronger expressed gene of the pair is greater than 1000; (3). the stronger expressed gene is marked "present" by the gene chip software. By these

criteria, 183 genes were identified after 8 and 14 days (corresponds to 1 day and 7days in the irradiated groups) vitamin A treatments accounting for about 2% of the genes represented on the chip. A large fraction of these vitamin A-responsivegenes, about 40% (77 genes), were unidentified ESTs lacking function information.

Vitamin A inhibited more transcripts than it induced, in contrast to radiation for which inductions outnumbered inhibitions. Genes related to epidermal differentiation and cellular metabolism were globally induced, while genes related to inflammation, energy genesis and cell movement were globally repressed.

As before, to avoid very high ratios between vitamin A and control sample, intensities less than 500 in the table were arbitrarily adjusted to 500. A similar trend of response to retinoids have been reported by other researchers, for example, retinoic acid induction of peptidylarginine deiminase (102), transglutaminase (103), and N-methyl-D-aspartate receptor (104).

Vitamin A regulated genes involved in epidermal differentiation

Several genes strongly up-regulated by vitamin A are known to be involved in epidermal differentiation, such as, involucrin, profilaggrin, protein-glutamine gamma-glutamyltransferase (i.e. transglutaminase), high sulfur protein B2E/B2F, which is a biomarker of the differentiation of hair matrix cells into hair fibers, small proline-rich protein (spr1), which is a marker for squamous differentiation, and peptidylarginine deiminase which produces a key component for the cornified envelope of terminally differentiated keratinocytes. All the above genes contribute to the formation of the cornified envelope of the epithelium, which is the hallmark of epidermal differentiation and confirms a major expected action of vitamin A in skin. The data presented here confirm that the retinoid's most profound effect is on the differentiation of epithelial cells. No evidence for action on other dermal skin cell types, such as, macrophages or fibroblasts, was found. Vitamin A also altered other keratinocyte structural genes, although the biological meaning of these alterations is not clear at this time. For example, cytokeratin type I gene expression was increased 5 fold at 8 days, but the increase did not persist to 14 days.

Proliferation and cell cycle related gene expressions altered by vitamin

Several pro-proliferation genes were down-regulated, such as, growth potentiating factor (D42148) and versican V3 isoform (AF072892), both of which regulate muscle cell proliferation, and rc_AI008639 which is about 60% homologous to *Drosophila* MAD in the c-myc pathway. DOC-2 p59 isoform was an exception, as it may have growth inhibitory effects on prostate epithelial cells, showed 3.8 fold vitamin A-mediated inhibition. Two cyclins with special relevance to skin, cyclin B and D2, showed 2.5 and 14 fold reductions by vitamin A treatment, respectively. Since overexpression of these cyclins causes hyperplasia in transgenic mice, their inhibition coupled with inhibition of other proliferation-related genes indicate the possibility of vitamin A-induced a slow-down of cell cycle progression and proliferation, most likely of the keratinocytes

Vitamin A regulates genes involved in energy metabolism

One striking alteration caused by the vitamin A was the inhibition of gene expression of the NADH-cytochrome b5 reductase and it's homologues at 8 days of

vitamin A treatment (not shown in the table). However, this inhibition did not persist to the end of 14 day vitamin A treatment regimen. Other energy-generating mitochondrial proteins --isovaleryl-CoA dehydrogenase, and acyl coenzyme A dehydrogenase were also inhibited. It appears that the addition of retinoid can lower energy-generation and may possibly a side effect of a slow down of the growth and proliferation rate of the affected cells.

A slowing down of energy generation by vitamin A is consistent with vitamin A's induction of terminal differentiation. Once the cells begin to differentiate terminally, their energy is directed toward making the cornified envelope and other cellular processes tend to slow. However, the lack of persistence of the inhibition of energy generation may mean vitamin A activated a feedback mechanism that brings the response back close to basal level by 14 days after the vitamin A is started.

Table 3. Genes that were most greatly altered by the addition of VA.

VA for	V0/C0 8 days	V0/C0 2 weeks	Description
1	8.7	2.0	L08228 N-methyl-D-aspartate receptor
2	4.9	1.9	X53581 long interspersed repetitive DNA with 7 ORFs
3	6.6	1.8	X05472 2.4 kb repeat DNA right terminal region
4	5.8	1.1	X54806 cytokeratin type I
5	4.1	1.7	rc_AA893219 =glycine methyltransferase
6	4.9	1.6	M13100c long interspersed repetitive DNA (L1Rn)
7	15.5	1.6	rc_AA800750 70% ~ to tyrosine aminotransferase
8	3.6	1.6	M21759 profilaggrin
9	2.9	1.4	L28818 RATINVO involucrin
10	3.9	1.6	J05022 peptidylarginine deiminase
11	5.2	1.6	cDNA clone rx03840 =histamine N-methyltransferase
12	4.1	1.5	M13100 long interspersed repetitive DNA (L1Rn)
13	6.4	1.3	L24207 testosterone 6-beta-hydroxylase (CYP3A1)
14	2.9	1.3	rc_AA799531 EST189028, ~ to a hypothetical gene
15	5.2	0.8	K02814 major acute phase alpha-1 protein (MAP)
16	3.3	1.2	rc_AA800651 EST190148, ~ to protein phosphotase 2
17	0.2	1.3	X16262 smooth muscle myosin heavy chain
18	0.3	1.2	D78359 drs (a gene down-regulated by v-src)
19	0.2	1.1	rc_AA800803 EST190300, ~ to no known gene
20	0.2	0.9	X83537 membrane-type matrix metalloproteinase *(UV up it in KC)*
21	0.3	0.9	rc_AA925246 ~ to cathepsin K, a cysteine proteinase
22	0.3	0.9	rc_AA799539 70% ~ mouse Nd1-L, relate to actin stabilization
23	0.1	0.9	J03179 D-binding protein, possible transcription factor
24	0.2	0.9	U26033 carnitine octanoyltransferase
25	0.2	0.9	rc_AA799745 EST189242, =CDK5 activator binding protein
26	0.3	0.8	X06916 p9Ka,~ to calcium-binding protein
27	0.2	0.8	rc_AI008639 60% ~ to Drosophila MAD (in c-myc pathway)
28	0.1	0.8	rc_AA892801 =eukaryotic translation elongation factor 2
29	0.1	1.0	AB012234 NF1-X1, nuclear factor 1 (transcription factor)
30	0.2	0.6	D12769 BTE binding protein (BTE=basic transcription element)

31	0.3	0.6	rc_AA892154 EST195957, up in quiescent fibroblast
32	0.2	0.5	AF072892 versican V3 isoform, muscle proliferation related
33	0.3	0.5	rc_AA893239 =2-hydroxyphytanoyl-CoA lyase
34	0.2	0.5	X63995 NTT, neurotransmitter transporter
35	0.3	0.7	U95178 DOC-2 p59 isoform (inhibits growth, may bind myosin)
36	0.3	0.4	D42148 growth potentiating factor, regulating VSMC proliferation
37	0.2	0.5	U62897 carboxypeptidase D precursor, may up inflammation
38	0.1	0.6	U68417 mito. branched chain aminotransferase
39	0.2	0.6	U83896 sec7B, secretory granules from Golgi
40	0.3	0.9	rc_AA799539, 5'~ to NADH-ubiquinone oxidoreductase
41	0.1	1	D16308 cyclin D2 (CLND2)

Abbreviations: Mito.- mitochondrial; VSMC - vascular smooth muscle cells

Vitamin A effects on transcription and protein synthesis

Since vitamin A promoted cells in the direction of terminal differentiation, it's reasonable for the cells to transcribe fewer RNAs and synthesize fewer new proteins. It's shown in the data that vitamin A inhibited several transcription factors, i.e., nuclear factor NF1-X1, BTE binding protein. By the same mechanism, *de novo* protein synthesis should be reduced, too. In the data, protein metabolizing enzymes, such as, uroporphyrinogen decarboxylase, carboxypeptidase D precursor, as well as, mitochondrial enzyme branched chain aminotransferase were all inhibited by the vitamin A.

Effect of vitamin A on invasion, migration and extra-cellular matrix genes

Usually malignant transformation is preceded by an increased expression of migration and invasion related proteins. In this study, one muscle migration and angiogenesis-related gene, versican V3 isoform precursor, was down-regulated by the vitamin A. vitamin A also inhibited membrane-type matrix metalloproteinase, which degrades components of extracellular matrix (collagen in skin) thus rendering cells potentially more invasive. In contrast, other types of matrix metalloproteinases were found to be UV inducible in human skin *in vivo* (105). These data suggest that *in vivo* vitamin A suppresses some transformation-related activities like invasion and migration. Since the regulated expression of cell invasion and movement gene products on endothelial cells is also central to the pathogenesis of various inflammatory processes, the retinoid's inhibition on those molecules may play a role in its anti-inflammatory effects as well.

Vitamin A's effect on inflammation-related gene expression

Agreeing with other research results, these data clearly show that the addition of vitamin A decreased several inflammation-related gene expressions, i.e., MHC class genes. One exception, however, was the two copies of the major acute phase alpha-1 protein (MAP), which correlates with acute inflammation, were increased by the vitamin A. Nevertheless, these data support the idea that the major effect of vitamin A was anti-inflammation, because more of inflammation gene transcripts were repressed than induced. At this time, it appears as if vitamin A's anti-inflammatory property might not

be attributable to its anti-oxidative property, since vitamin A repressed inflammation-related genes but did not affect the expression of oxidative stress-related genes.

Figure 15 shows changes in the number of vitamin A-affected genes in each functional category at day 8. Comparing the number of genes responding to the vitamin in comparison to genes expressed in control, it can be seen that vitamin A affected genes mostly in the categories of 1) epidermal differentiation, 2) inflammation, 3) nucleic acid metabolism and 4) cytoskeleton/matrix/adhesion. vitamin A did not significantly affect the number of genes in the categories of 1) apoptosis, 2) cell cycle, or 3) stress response.

Efforts were made to find similar expression patterns among genes within categories. In many categories, similarities were not evident, perhaps due to the complexity of the categories. However, in several groups, a majority of genes followed the same trend of regulation, indicating a possibility of global regulation. vitamin A led to global induction in epidermal differentiation and cellular metabolism categories, while global repression was seen in inflammation/immune and cytoskeleton/matrix/migration categories.

Pathways not affected by the vitamin A

Initially it was expected that vitamin A might exert its effect by synergistically increasing radiation-induced apoptosis, effectively killing more cells and reducing the chance of passing DNA damage to descendant cells, however, the data did not support

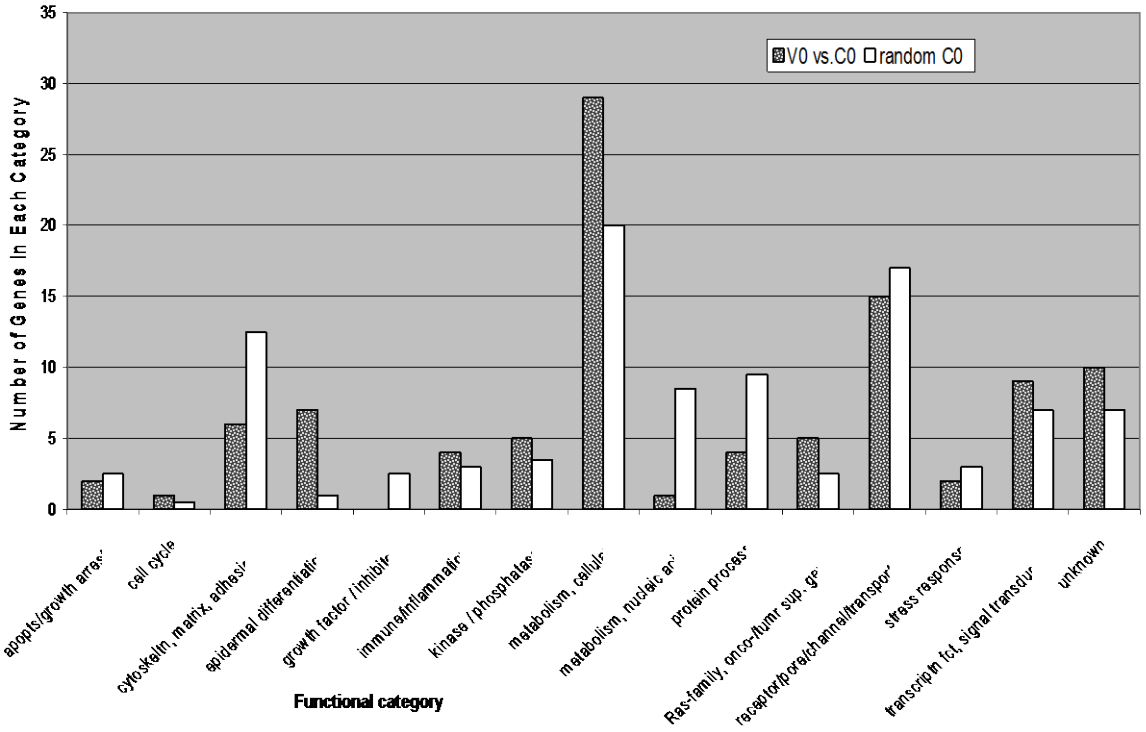


Figure 15. Functional Grouping of the Genes that were Most Greatly Affected By 8 day vitamin A Treatment.

this idea. The data showed no alteration of apoptosis-related gene expression after either 8 or 14 days of vitamin A treatment. Neither did vitamin A alter DNA repair gene transcripts at the time points studied. Cytokines, including interferons and interleukins, transforming growth factor, tumor necrosis factor alpha, generally did not pass the screening criteria used in this study. However, one exception was tumor necrosis factor converting enzyme (AJ012603, TACE) which showed a 2.7 and 1.3 fold repression after 8 and 14 days vitamin A treatment, respectively.

In summary, this current study discovered that retinyl acetate in the rat diet could alter gene expression pathways in the skin as summarized in Figure 10. The figure shows vitamin A inhibited epidermal proliferation, inflammation, and cell migration/invasion, while promoted epidermal differentiation. The figure also shows vitamin A may increase immuno-surveillance of the treated rats, based on current data and the studies by Sailstad and his co-workers (1995, 2001).

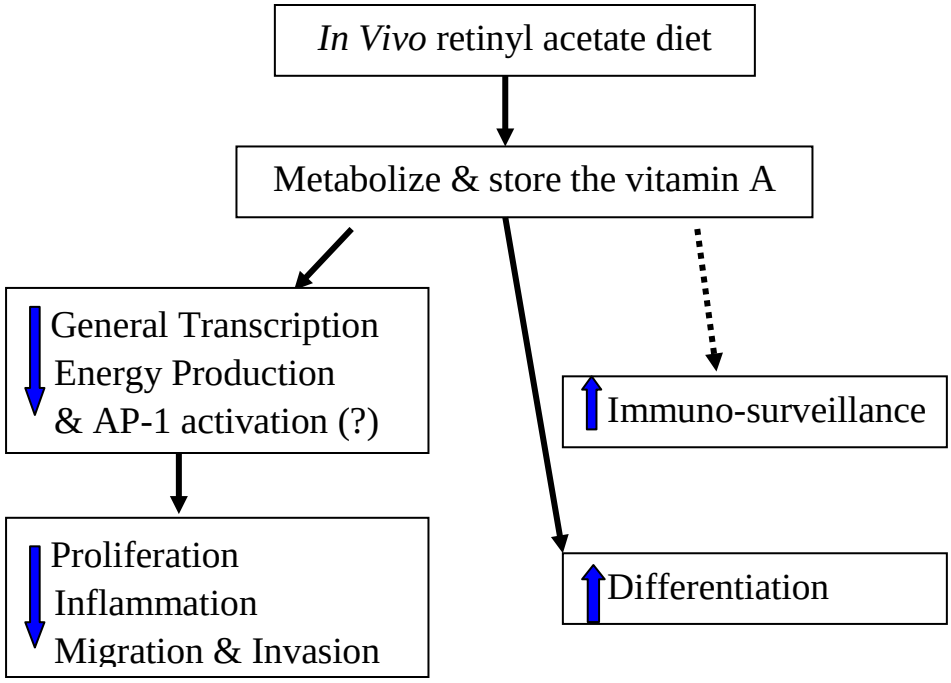


Figure 16. Schematic representation of *in vivo* retinyl acetate's effects on the rat skin. The actions were identified in this study, or linked by steps from other researchers as marked by dashed arrow.

INTERACTIONS BETWEEN RADIATION AND RETINYL ACETATE EFFECTS:

In the previous chapters the radiation effects and the retinyl acetate effects on the rat skins were discussed separately. In *in vivo* conditions, these actions may interact with each other in a way that is not predictable from examining the 2 treatments separately. To discover interactions between electron radiation and vitamin A effects, skin samples that had received both electron radiation and vitamin A were compared with other samples that had only one or neither of the treatments. Theoretically effects may be interactive or

independent. Interactive effects would include 1) reversal effects, i.e. in opposite directions or 2) additive and 3) synergistic effects, i.e. in the same direction. Independent effects would occur when the individual effects of both treatments alone were additive when combined in the same animals. These various interactions will be discussed separately in the following sections, with emphasis on positive and negative interactive effects, and least emphasis on the independent actions.

Before going into a detailed study of electron radiation and vitamin A interactions, a functional study of the vitamin A plus electron radiation affected genes is described below, so as to give a general view of their actions. To do this, the most greatly affected 100 genes by 9 Gy electron radiation plus vitamin A (vitamin + 9Gy vs. control) were fitted into the fifteen functional categories, then the number of genes in each category was plotted as in Figure 8. As control, 100 randomly picked up genes from the expressed control sample were also assigned to functional categories and plotted in the same chart, together with vitamin A alone (vitamin only vs. control) or the 9 Gy electron radiation alone (9Gy vs. control) as shown in Figures 4 & 5, for reference purpose. The list of the genes that were used to generate Figure 8 is attached in Appendix.

Figure 17 gives a general view of the functional categories of the genes that were strongly regulated in the electron radiation+vitamin A sample (vitamin + 9Gy) in comparison to control control at 1 day post 9 Gy electron radiation (i.e., 7 days after the vitamin A treatment). The chart shows how the 100 most strongly affected genes by 9 Gy electron radiation plus vitamin A (vitamin + 9Gy) fell into the various categories. The categories most affected were 1) epidermal differentiation, 2) cellular and nuclear metabolism, and 3) cytoskeleton/matrix/migration, implying the involvement of these three cellular pathways in the interacting response to vitamin A and electron radiation treatments. By comparing the vitamin + 9Gy vs. control to that of 9Gy vs. control and vitamin + 9Gy vs. control, the figure shows vitamin A tended to reverse electron radiation effects in 1) protein process, 2) cytoskeleton/matrix and 3) stress response genes, while in the remaining categories, vitamin A did not significantly alter the number of genes affected by electron radiation.

Reversal Effects Of Retinyl Acetate On Irradiated Skins

One of the most important goals of this study is to identify genes whose electron radiation-altered expression could be reversed by the retinyl acetate. Our data showed some interesting results in revealing reversal effects of vitamin A versus electron radiation. Because the experiments were carried out *in vivo*, the results would be more readily applied to human models, in regard to exploring mechanisms and developing potential therapeutic treatments.

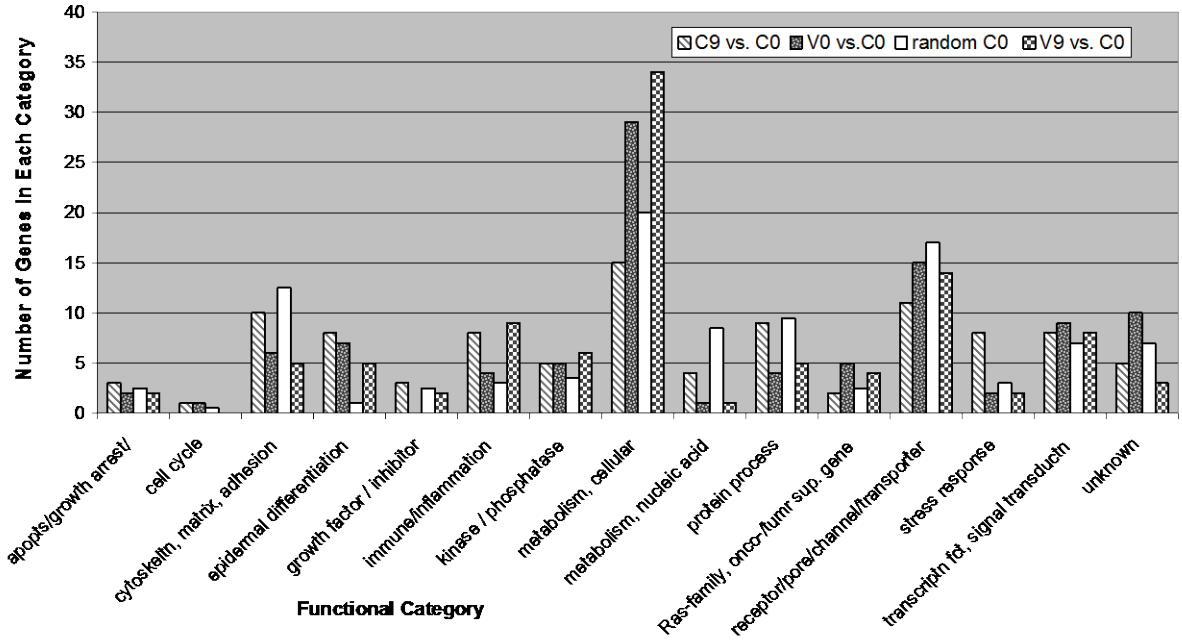


Figure 17. Comparison of the functional groups that were affected by electron radiation and vitamin A at 1 day post-electron radiation. The 100 most greatly altered genes for each treatment including 9 Gy electron radiation plus vitamin A, 9 Gy electron radiation alone or vitamin A alone versus no treatment control control were assigned into each of the 15 functional groups. The percentages of expressed genes on the gene chip in the various categories in the control (control) were included in the chart for comparison to show how genes would distribute if they were randomly selected.

Sixteen genes showed at least a 2.5 fold induction (or reduction) by 9 Gy electron radiation (9Gy) versus control, and at the same time showed at least 2.5 fold change in the opposite direction, i.e., repression (or induction) in the 9 Gy electron radiation plus vitamin A (vitamin + 9Gy) versus control sample. These genes were listed in Table 4. For clarity, the expressions were represented in the table by ratios over control control, i.e., the ratio for genes in controls was set to 1, and the expressions of other genes were replaced by the fold change relative to controls. To avoid extreme and illogical ratio values, all expression values less than 500 were set to 500.

Different expression patterns were found as detailed in Table 4. In Group #1, gene expressions were induced by electron radiation alone and were or were not repressed by vitamin A alone (subgroup A and B, respectively), while the electron radiation inductions were reverted (reduced) by the co-existence of the vitamin A. There was only one gene (inositol tri-phosphate receptor subtype 3, L06096) in Group #2, showing a vitamin A reversal effect as vitamin A induced the gene expression that was inhibited by the radiation. Among the genes that showed the greatest reversal of electron radiation effects (9Gy) by vitamin A treatment (vitamin + 9Gy) were: keratin 14 (K14), MRP-14 (intracellular calcium-binding protein gene), and an EST rc_AA957003 which is homologous to intracellular calcium-binding protein MRP 8. As a general rule, the magnitude of such reversals was greatly diminished by 7 days post-electron radiation.

Examples of genes that showed the greatest vitamin A reversal of electron radiation effects

Several genes in Table 4 may have special implications, they are described below:

♦Keratin 14 (K14) is a basal cell marker marking proliferating cells (i.e., basal cells) of the epithelium, it is not expressed in the dermal cells of the epithelium. It is strongly overexpressed in human squamous cell carcinomas (electronic-PCR data from NCBI: National Center for Biotechnology Information). In this experiment, k14 showed a very strong induction by 9 and 18 Gy electron radiation at 1 day post-electron radiation, and the addition of vitamin A in the equally irradiated samples reversed the induction almost to the control level. This electron radiation induction could still be seen in the 7 day post-electron radiation sample, although to a much lesser extent.

♦Also of great interest are the two genes: MRP-14 (intracellular calcium-binding protein gene), and EST rc_AA957003 which is homologues to intracellular calcium-binding protein MRP 8, the two gene showed strong electron radiation induction and an 8-day vitamin A treatment reversed the expressions almost back to control levels. Unfortunately the probe for MRP8 gene was not on the gene chip, however through a BLAST search on the NCBI website (www.ncbi.nlm.nih.gov), the EST rc_AA957003 showed 95% similarity to MRP8 gene. It's known that MRP14 and MRP8 are both calcium related, they heterodimerize to form an intracellular calcium-binding protein. The binding could act as a buffer to reduce the free intracellular calcium ions, which in turn, may suppress differentiation of the epithelium. Recently this pair of genes has been found to be injury-regulated and involved in skin carcinogenesis (106, 107) in mouse models, suggesting a possible significance of their inhibition in a cancer prevention process. In our experiment, the electron radiation-induced Ca^{2+} -binding gene expressions were repressed by the addition of vitamin A, showing vitamin A could at least partly restore the epithelial calcium balance that was disrupted by the electron radiation. Thus the data suggest intracellular calcium regulation may be involved in vitamin A reversal of electron radiation effects on rat skin.

Originally these two calcium-binding genes were identified in connection with inflammation, as the name migration inhibitory factor-related protein (MRP) implies. *In vivo*, MRP8 and MRP14 form a heterodimeric complex in the cytosol of monocytes and especially neutrophils. More detailed study revealed MRP14 is a novel down-regulatory molecule of macrophage activation (108). Since macrophages are anti-neoplastic surveillance cells, the down-regulation of their activation may contribute to immuno-suppression after electron radiation, while immuno-suppression, sometimes hypothesized as a basis for cancer induction in skin, could let neoplastic cell escape the immuno-surveillance and progress to cancer cells. At 7 day post-electron radiation, the electron radiation induction and the vitamin A reversal effects of these genes became less striking than that at 1 day post-electron radiation, nonetheless the reversal effects were still present.

Vitamin A and electron radiation reversely regulated inflammation and possibly immune suppression

By visual inspection of the rats at 1 day through 7 days after the electron radiation, the vitamin A supplement clearly inhibited the erythema in comparison to the electron radiation-only rat skins. In addition, it was observed that the electron radiation-related skin desquamation also appeared later and less severely in the vitamin A supplemented rats in comparison to electron radiation alone. These observation demonstrate that vitamin A inhibited the inflammation caused by the electron radiation. See Figure X panel Y for the irradiated skin with or without vitamin A supplement.

In agreement with the visual observations, several gene transcripts related to inflammation showed vitamin A inhibition of electron radiation induction, such as beta defensin-2 (AF068861), RT1.Au gene (X82669), MHC class II-associated invariant chain (X14254), complement protein C1q beta chain (X71127), MHC class II A-beta RT1.B-b-beta, as well as MRP14 and a MRP8 homologue as described above in Chapter 6.1.1. It's not surprising to discover that after 8 and 14 days vitamin A treatment, different inflammation-related transcripts were inhibited. Based on the visual and gene expression studies, our *in vivo* data clearly demonstrated vitamin A could repress and delay the electron radiation-induced skin inflammation response.

Since macrophages express high MHC II molecules, the induction of several MHC II gene expressions as seen in our experiment may point to the reversible infiltration of macrophages to the irradiated areas. In human skin UV induction of the infiltration of MHC II-expressing macrophages has been found to suppress cellular immune responses in the skin (109). Similarly, the electron radiation-induced MHC II expression may participate in the immuno-suppression responses in the rat skins in this study. Furthermore, as discussed in 6.1.1, electron radiation-induced MRP14/MRP8 may also contribute to immuno-suppression by down-regulating macrophage activation. On the other hand, dietary retinyl acetate has been shown to boost murine immuno-surveillance, which tends to support the above idea that vitamin A may have reverted electron radiation-induced immuno-suppression.

Vitamin A and electron radiation reversely regulated epidermal proliferation

The expression data show vitamin A repression of several electron radiation-stimulated epidermal proliferation-related transcripts, for example, keratin 14 (D63774), and MEGF5 (AB011531) which has multiple epidermal growth factor-like motifs, and an EST (rx04836) which has homology with mouse squamous cell carcinoma (SCC) antigen. These changes support the idea that vitamin A acted counter to electron radiation-induced epidermal proliferation, which agrees with many other studies on the individual effects of electron radiation and retinoids on skin, i.e., electron radiation causes hyperproliferation for weeks after exposure, while retinoids have growth inhibitory effects.

Oxidation-related genes might be involved

Among the genes that were strongly reversely regulated by the vitamin A and electron radiation, was an oxidation-related transcript: urate oxidase (uricase, X13098). Another probe (J03959) for this gene also showed a similar reversal pattern but to a lesser

Table 4

~~24 hour post IR~~					~1 week post IR~		
C9-I	C18-I	V0-I	V9-I	Gene Description	C9-II	V0-II	V9-II
Group 1. RA reverses Rad induction							
G1-A. RA alone does not alter the gene expression							
12	4.6	0.9	1.7	D63774 RATRETK keratin 14,	2.2	1.7	1.6
46	59.2	1.5	3	L18948 intracellular calcium-binding prt (MRP14'	3	1.7	2.8
16	8.4	1.5	8.4	M13949 cellular retinol binding protein II (CRBPII	3.4	0.4	2.3
22	18.3	1.7	9.9	clone rx04836, ~ mouse SCC antigen 2	4	1	3.2
3.7	4.4	0.6	0.7	rc_AA957003, ~ to MRP8	1.2	1	1.1
9.8	9.6	1	1	AF068861 beta defensin-2	1	1	1
5.8	2.2	1	1	X82669 RT1.Au gene	1	1.2	1
2.8	1.5	1	1	J05495 (macrophage) Gal/GalNAc-specific lectin	1.3	0.8	0.9
5.6	9.2	1	1	X13098 urate oxidase (uricase)	1	1	1
4.8	8.4	1	1	AF017393 cytochrome P4502F4 (CYP4502F4)	1	1	1
1.7	2.6	1	1	AB011531 MEGF5, with multiple EGF-like motifs	1.2	0.8	1
G1-B. RA alone also inhibits the gene expression							
1.5	0.6	0.3	0.6	K02815 MHC RT1-B region class II A-alpha	2.6	0.9	1.1
2	0.8	0.3	0.4	X14254 MHC class II-associated invariant chain	2.7	1.5	2.1
2.8	2.4	0.6	0.6	X71127 complement protein C1q beta chain	1.9	2.8	1.9
1.2	2.1	0.2	0.3	U36992 cytochrome P450 Cyp7b1	1	1	1
Group 2. RA reverses IR inhibition							
0.6	0.6	0.9	0.9	L06096 inositol tris-P receptor subtype 3 (IP3R-3'	0.5	0.9	1.5

degree (data not shown). This implies vitamin A might have some anti-oxidation properties that could help to reduce the oxidizing stress of electron radiation on the rat skin. It was not surprising to find such an effect on a gene that could be part of the oxidative stress reaction, although the effect was only present at 1 day post-electron radiation and not at 7 days. Another gene, cellular retinol binding protein II (CRBPII, M13949), also clearly demonstrated vitamin A reversal of electron radiation induction. The electron radiation induction of CRBPII was likely the result of the depletion of the retinoid in the skin after electron radiation damage, but it is not clear if the depletion is caused by oxidative stress.Table 4. List of the genes showing vitamin A reversal of electron radiation effects

Gene groups that did not show vitamin A reversal of electron radiation effects

Several gene categories failed to show any significant reversal effects between electron radiation and electron radiation plus vitamin A samples (9Gy vs. vitamin + 9Gy). They include oncogene/tumor suppressor genes, kinase/phosphatase genes, and receptor/pore/channel/transporter genes. This result supports the notion that oncogene activation is a late event, possibly occurring during the progression phase of tumor formation, rather than at the initiation and promotion phases.

Genes regulated by vitamin A and electron radiation toward the same direction

Genes were sought that were regulated by the radiation and the retinoid in the same direction (including additive or synergistic effects). To do this, all the genes in electron radiation and vitamin A samples that showed a greater than 2.5 fold change (induction or repression) against their control levels were assembled into a list. As before, the listed genes had to meet 2 other criteria, i.e., the stronger gene in each comparison pair had to be greater than 1000 in expression level and at the same time be a "P" (present) according to Affymetrix software.As anticipated, not many genes turned up. Several epidermal differentiation related genes were up-regulated by both electron radiation and vitamin A, some even showed additive or synergistic effects in the vitamin A+electron radiation samples, i.e., vitamin + 9Gy at 1 day or 7 days post-electron radiation. Examples include gamma-glutamyl hydrolase (U38797) precursor protein, gamma-glutamyltransferase (M57263), high sulfur protein B2E/B2F (AB003753), and cytoskeleton type I (x54806). The results show both electron radiation and vitamin A promote epidermal differentiation, thus confirming the results obtained from studying their individual effects on the skin.

Table 5. Genes that were regulated toward the same direction by electron radiation and vitamin A

	C9/C0	C18/C0	V0/C0	V9/C0	Gene Description
	1 day post-eRad				
1	0.3	0.4	0.2	0.2	AA684960 5'~ NADH-ubiquinone oxidoreductase
2	0.3	0.4	0.1	0.1	AA685376 5'~ NADH-ubiquinone oxidoreductase
3	0.2	0.1	0.1	0.1	J03179 D-binding protein
4	0.3	0.2	0.2	0.5	X63995 NTT
5	0.3	0.4	0.2	0.4	U20796 nuclear receptor Rev-ErbA-β
6	0.3	0.4	0.4	0.2	S69206 protease 1
7	4.9	2.5	5.9	5.7	X54806 cytokeratin type I
8	9.7	8.4	4.8	6.0	M57263 γ-glutamyltransferase
9	8.4	3.1	4.3	19.7	U38379 γ-glutamyl hydrolase precursor
10	20.0	12.2	68.3	21.2	AB003753 high sulfur protein B2E/B2F
11	5.5	3.6	2.7	8.0	M58308 histidase
12	4.4	3.2	3.2	4.5	AF068860 beta defensin-1
13	13.1	8.9	7.2	13.6	AF093536 beta defensin-1 (BD-

					1)
7 days post-eRad					
1	3.0	N/A	2.8	3.5	U78977 putative ATPase
2	2.5	N/A	3.0	2.4	D00680 plasma glutathione peroxidase
3	3.3	N/A	3.6	4.8	AF061971 palmitoyl-protein thioesterase2
4	4.1	N/A	4.0	6.1	M61142 metalloendopeptidase

N/A: data not available for the comparison.

Three transcription factors D-binding protein(J03179), NTT (X63995), and nuclear receptor Rev-ErbA-β (U20796) were repressed by both electron radiation and vitamin A. Together with their inhibition of two transcripts with 5' similar to mitochondrial electron transport chain component NADH-ubiquinone oxidoreductase (AA685376 and AA684960), data then suggests in addition to the vitamin A inhibition of growth (as discussed in Chapter 5), electron radiation may stop or delay growth of some cell types. Another transcript, hitidase (M58308), showed synergic induction at 1 day and additive induction at 7 days. But the biological meaning of these transcript expression differences is not yet clear.

Vitamin A and electron radiation's independent actions on the skin

Except reversal and same trend regulations by electron radiation and vitamin A, electron radiation and vitamin A may also have actions that are independent to each other treatment. Genes that were regulated independently were only altered by one of the two treatments, however, in the combined treatment sample, the genes were regulated as if only one treatment were present. One example is that several myosin-related transcripts showed similar degree of electron radiation induction in 9Gy and vitamin + 9Gy (i.e., 9 Gy electron radiation with or without vitamin A), while vitamin A alone had almost no effect. Other example is the gene for an invasion-related protein membrane-type matrix metalloproteinase (X83537), which had been repressed by vitamin A by about 10 fold irrespective of the presence of electron radiation. Since such independent actions are not the main focus of this study, and many of them have been mentioned in the studies of individual electron radiation or vitamin A actions, so here such actions will only be stated briefly.

Interactions between electron radiation and vitamin A based on data from the gene expression microarray study suggest that vitamin A can inhibit electron radiation-induced proliferation and inflammation responses, and some fibroblast or macrophage migration/invasion responses as well. Furthermore, vitamin A promoted epidermal differentiation on top of the electron radiation-stimulated abnormal epidermal differentiation, thus possibly protecting the skin from cancer progression by promoting the death of potentially electron radiation damaged cells. This idea is depicted in figure 12.

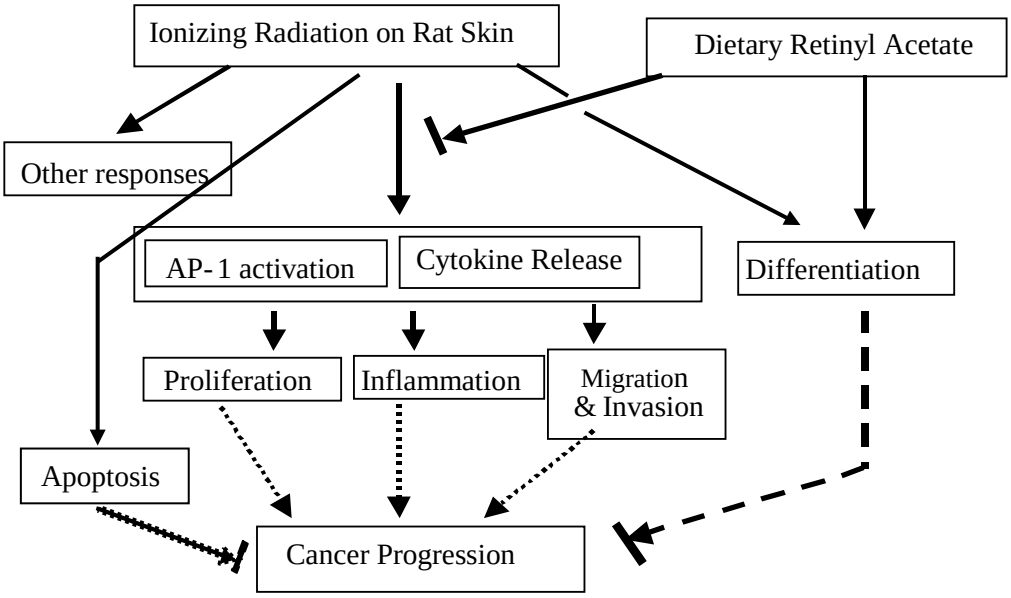


Figure 18. Schematic representation of retinyl acetate's reversal effects on electron radiation actions on rat skin, and some possible pathways in cancer prevention.

DISCUSSION:

Results from the present study are consistent at the gene expression level with prior speculations about the possible mechanism of retinoid's inhibition of skin cancer promotion and progression, that is, the retinoid was actively blocking the expression of genes associated with increased proliferation and inflammation following irradiation and accelerated the expression of genes associated with epidermal differentiation. Unfortunately, there are many seemingly contradictory reports on retinoid action on skin or skin cells, for example, retinoid induced proliferation in human skin, but inhibited differentiation in human keratinocytes (110). Such controversy may be explained, at least in part, by inconsistencies in the design of conflicting experiments in addition to the complex ways that retinoid may alter cellular differentiation. Not only are cellular and microenvironmental factors, such as different cell lines, different routes of administration (i.e., topical, dietary, etc.) important, but also different actions may arise based on differences in the dosage or chemical forms of the retinoid. In hopes of avoiding these confusions, we adopted an *in vivo* rat skin model and used naturally-occurring retinoid, i.e., retinyl acetate, at a non-toxic concentration as dietary supplement.

Rat keratinocyte cultures were used at the beginning of this study, but the cultures showed high levels of spontaneous transformation when tested by the soft agar assay for anchorage-independent growth (data not shown), in spite of these cells being within the first few passages of primary keratinocytes obtained from newborn rats. The *in vivo* model has overcome this problem inherent in the *in vitro* system.

Recent years have seen some rapid advances in the biotechnology field, one of the most fascinating of which is the availability of nucleotide microarray (gene chip) technology to the general research public. The gene chip, manufactured by Affymetrix Inc., can produce a large amount information about gene expression in a much shorter time than conventional "one-gene-one-experiment" hybridization methods. The Affymetrix GeneChip® probe array is manufactured through a unique and robust

process— a combination of photolithography and combinatorial chemistry to synthesize the probes "on-site" on a small chip of quartz of mere 1.28 square centimeter dimensions. With a minimum number of synthesis steps, GeneChip® technology produces arrays with tens of thousands of different oligonucleotide fragments (probes) packed at extremely high densities at accurately known locations. Only about four years ago we first started to explore this vitamin A reversal of electron radiation effects, when the best technology was cDNA macroarrays with only 588 gene probes on a nylon membrane (Clontech, CA), and that proved to have inherently poor reproducibility (data not shown). Without the newer Affymetrix microarray technology this thesis work could not have been done with such a broad array of genes (more than a dozen times more genes than the older nylon membrane arrays), and with at least a factor of 2 better reproducibility.

However, even though gene chip technology has many advantages in studying gene expression patterns, data mining from the huge amount of gene expression data generated by gene chip hybridization is still a developing field. The foundation of gene chip data analysis lies on the comparison of the hybridization intensities of different genes across samples with or without treatment. The availability of large number of genes on the microarrays made it possible to study single genes and large groups of genes with similar functions independently, thus providing a far more complete gene expression profile that could be achieved without this technology. The fast spread of gene chip technology has stimulated the development of many software techniques specifically designed to analyze the data, for example, Clustering and Gene Tree (Stanford University, CA), GeneSpring (Silicon Genetics, CA), self-organizing maps (SAM, available from many sources), Affymetrix Microarray Suite 5.0/Data Mining Tool (Affymetrix, CA), Spotfire DecisionSite (Spotfire Inc., MN), etc. Some of the software is freely available on-line, while other software is commercial. Some are just for image analysis, like Imagene 5.0 from BioDiscovery Inc. (CA). A careful review of the various methods suggests that the clustering and gene tree analyses are the most informative methods for seeking genome-wide patterns. Other software may provide some additional features, especially, visualization tools including Venn diagrams, bar charts, scatter plots, chromosome display and dendrograms, etc., such as GeneSpring provides. In our experience, Microsoft Excel is a useful tool to single out differentially displayed genes, although it doesn't support any of the visualization functions.

Studies by others have revealed that one of the main mechanisms of the retinoid effect lies in its inhibition of AP-1 (activator protein-1) activation, possibly through indirect binding of c-jun/fos to vitamin A-RAR (111). It is also interesting that AP-1 transactivation is required for tumor promotion (112). In addition, the Ets and AP-2 sites also appear to be critical and collaborate to regulate epidermal specificity *in vivo* (113), and that protein kinase C (PKC) activation is a master gene for controlling the epidermal differentiation process (114). The AP-1/AP-2 inhibition is essential for the retinoid's growth suppression and differentiation-inducing effects. In our data, many of the vitamin A-regulated genes do have either AP-1 or AP-2 sites on their promoters, for example, involucrin has two functional AP-1 responsive sites on its promoter, profilaggrin and transglutaminase and small proline-rich protein (spr) each has at least one AP-1 site. vitamin A inhibited the expression of matrix metalloproteinase as seen in our data, possibly through the repression of AP-1 activation (115). Therefore, the differentiation process induced by the addition of vitamin A in our study could be mediated through the

activation of PKC and the downstream AP-1/AP-2 transcription factors. However, this AP-1/AP-2 theory fails to explain why in some other circumstances, the retinoid may have opposite effects on the transcription of the same differentiation genes.

Another well documented yet important pathway involved in epidermal differentiation is the calcium pathway, where an increase of intracellular free calcium ion concentration favors differentiation. Our data showed vitamin A inhibited 2 calcium-binding-related gene transcripts: dihydropyridine-sensitive calcium channel alpha-2, and the metastasis-inducing p9Ka whose protein product is located on cytoskeletal elements and is similar to calcium-binding protein (it also stimulates differentiation of epithelial cell into myoepithelial cells). These two transcripts both favor differentiation of keratinocytes, because they would tend to reduce intracellular free calcium. Since intracellular calcium level affects the differentiation status of the epithelial cells, the results of the current study imply that both the calcium pathway and the AP-1 pathway may be involved in the observed rat skin reactions.

Interestingly, PKC activation triggers calcium ion influx in epidermal cells **(**Eckert et al., 1997), which may partially explain the differentiation-inducing effects of the activation of PKC. Furthermore, this observation links the possible AP-1/AP-2 mediated differentiation and calcium involvement to the general control of the PKC activation.

The present study suggests retinyl acetate's inhibition of ionizing radiation effects on rat skin is modulated through epidermal proliferation and inflammation; however, there may be other mechanisms responsible for the retinoid's cancer preventive action. Other reports show that in addition to induction of inflammation, ionizing radiation also has immuno-suppressive effects in mouse skin models (116). This immune suppression was found to be associated with the release of cytokines, a defect in antigen presentation, induction of suppressor T cells and suppression of contact hypersensitivity (117). The immune suppression sounds paradoxical to the fact that ionizing radiation also induces inflammation, however, a closer check reveal the two end points differ at protein, mRNA and cytogenetic levels (117). This phenomenon explains why some tumor implants grow much easier in UV-irradiated mouse skin than in non-irradiated control skin (118), linking host immuno-surveillance to the survival of malignant cells. Interestingly, dietary retinyl acetate can reverse the ionizing radiation caused immuno-suppression in rodent models as measured by contact hypersensitivity (117). Based on this observation and other similar studies, immune-suppression is likely to be relevant to skin cancer development (119). However, the gene chip data in this present study was not informative regarding this immuno-suppression effect, perhaps due to the dilution effects of the lymphatic response among the epidermal responses. Nevertheless, vitamin A reversal of electron radiation-caused immuno-suppression remains a possible mechanism of vitamin A-induced prevention of electron radiation actions on rat skin.

As a late effect of high doses of ionizing radiation injury, many exposed tissues including skin, lung, kidney, liver may develop fibrosis as a result of abnormally increased fibroblast proliferation and differentiation, as well as, collagen secretion (120) reviewed by Rodemann (121). In a recent study, ionizing radiation of 1.0 GeV/nucleon ^{56}Fe ions induced skin fibroma on rat skin, which could be partially inhibited by dietary retinyl acetate supplement (1). Fibroblasts (especially myoblasts) play a major role in the

formation of such fibrotic lesion in response to ionizing radiation, while inflammation also participates in this process through macrophage activation and infiltration of inflammatory cells, plus other cellular responses. Our present study didn't last long enough to establish whether rat skin fibrosis or fibroma induction might have responded to the retinoid, however the gene expression data at 7 days post-electron radiation did show alterations potentially relevant to fibroblast migration and proliferation, i.e., strong induction of several types of myosin transcripts. Despite about 50% of the ionizing radiation-induced fibromas being inhibited by dietary vitamin A (1), our present study showed only one myosin transcript in the vitamin A plus electron radiation sample was a vitamin A reversal of electron radiation induction, while most of the other electron radiation-induced myosins were vitamin A independent, suggesting that vitamin A may not inhibit electron radiation induced fibrosis by inhibiting the expression of myosin transcripts. Besides muscle cells, the myosins could also be produced by fibroblasts or macrophages or other cells of dermal origin, thus possibly connecting electron radiation to the up-regulation of cell migration, or lymphatic cell infiltration, or inflammation. Even though vitamin A barely reverted the electron radiation-induction of myosins, vitamin A's anti-inflammatory properties may contribute to the inhibition of fibrosis.

OTHER INFORMATION AND MATERIALS:

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1.0 Burns, Fredric J., Zhao, Ping, Xu, G., Roy, Nirmal, Loomis, Cynthia. Fibroma Induction in Rat Skin Following Single or Multiple Doses of 1.0 geV/nucleon 56Fe Ions from the Brookhaven Alternating Gradient Syncrotron (AGS). Physica Medica 17:194-5 (2001).

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5. Students supported by the Project:

1. Shuaili Chen, Ph. D. 2003

2. Ping Zhao, M.S., 2001

3. Guijuan Xu, M.S., 2001

4. Feng Wu, Ph. D., Postdoctoral Fellow, 2002

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